

# Quantum control for quantum information transfer mediated by topological edge states

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(Dated: May 2026)

We study quantum state transfer in one-dimensional topological chains based on the Su-Schrieffer-Heeger (SSH) model, comparing two distinct paradigms: domain wall-mediated transfer and counterdiabatic (CD) driving. First, we numerically reproduce a domain wall transfer protocol, achieving fidelities  $f > 0.999$  for symmetric two-domain configurations with  $L = 11$  sites. We investigate the effect of asymmetric domain wall placement, demonstrating that the fidelity is constrained by the exponential sensitivity of the effective couplings to the domain lengths—a limitation of the static Rabi protocol rather than of topological protection itself. We further explore shortcut-to-adiabaticity (STA) pulse protocols and find that no pulse optimization can overcome the coupling asymmetry that governs the transfer phase. Second, we present an extensive analysis of a hybrid analog-digital protocol for nonadiabatic edge-state transfer, where the SSH Hamiltonian is supplemented with approximate counterdiabatic interactions derived from adiabatic gauge potentials and optimized via variational quantum circuits. Third, motivated by the CTAP (Coherent Tunnelling Adiabatic Passage) formalism, we show analytically that the coupling asymmetry can be overcome by replacing the static Rabi protocol with a dynamic counter-intuitive pulse sequence that exploits the topological dark state of the effective three-level Hamiltonian. We discuss extensions to two-domain wall systems, showing that the even-state parity constraint imposes fundamental limitations on the standard CTAP scheme.

## I. INTRODUCTION

Quantum state transfer—the coherent transport of quantum information between distant nodes of a quantum network—is a fundamental primitive for quantum computing and communication architectures [1–3]. In solid-state platforms such as quantum dot arrays, superconducting circuits, or cold-atom lattices, the transfer must contend with disorder, decoherence, and the difficulty of precisely engineering long-range couplings.

A promising approach exploits the *topological protection* of edge states in one-dimensional lattice models [4, 5]. The Su-Schrieffer-Heeger (SSH) model [6, 7]—a paradigmatic example of a symmetry-protected topological insulator in class BDI [8]—supports exponentially localized zero-energy edge states when the system is in the topological phase. These states are robust against local perturbations that respect the protecting chiral symmetry, making them attractive candidates as quantum information carriers.

J. Zurita, C. E. Creffield, and G. Platero [9] demonstrated that chains comprising multiple SSH domains separated by *topological domain walls* can mediate fast, high-fidelity quantum transfer. The key mechanism is an exponential speed-up arising from the multi-domain structure: by distributing  $N$  domain walls along the chain, the transfer time scales as  $(w/v)^{\ell/2}$  per domain (where  $\ell$  is the domain length and  $v/w$  the hopping ratio) rather than  $(w/v)^{L/2}$  for the entire chain. This exponential compression of the transfer time, combined with the topological robustness of the boundary states, provides a powerful paradigm for quantum information transport.

In this work, we present a study on topological quantum transfer in SSH chains. We focus on two-domain

( $N = 2$ ) SSH chains and address three questions:

1. Can the transfer protocol of Ref. [9] be faithfully reproduced numerically, and what are the optimal parameters?
2. How does the transfer fidelity degrade when the domain wall is displaced from the chain center, breaking the symmetry between the two domains?
3. Can shortcut-to-adiabaticity (STA) pulse protocols [10–12] improve the transfer performance, particularly in asymmetric configurations?

Additionally, we provide a comprehensive analysis of an alternative approach: the hybrid analog-digital protocol recently proposed by S. V. Romero, X. Chen, G. Platero, and Y. Ban [13], where nonadiabatic transfer is achieved through counterdiabatic (CD) driving rather than topological domain walls. We compare both paradigms in terms of their physical mechanisms, scalability, and robustness.

Finally, motivated by the CTAP framework of A. D. Greentree, J. H. Cole, A. R. Hamilton, and L. C. L. Hollenberg [14], we show how the limitations of the static Rabi protocol in asymmetric configurations can be overcome by means of a dynamic counter-intuitive pulse sequence that exploits the topological dark state of the effective three-level Hamiltonian.

The remainder of this document is organized as follows. Section II introduces the SSH model, its topological properties, and the domain wall construction. Section III describes the transfer protocol and presents our reproduction of the main results of Ref. [9]. Section IV analyzes the effect of asymmetric domain wall placement. Section V investigates STA pulse protocols. Section VI

presents a detailed exposition of the hybrid analog-digital counterdiabatic protocol. Section VII provides a systematic comparison of the domain wall and counterdiabatic approaches. Section VIII introduces the CTAP framework as a route beyond the static Rabi limitation, analyzes the topological dark state, and discusses extensions to two-domain wall systems. Section IX summarizes our findings and outlines future directions.

## II. THE SSH MODEL AND DOMAIN WALLS

### A. The Su-Schrieffer-Heeger Hamiltonian

The SSH model describes a one-dimensional tight-binding chain with alternating hopping amplitudes  $v$  (intracell) and  $w$  (intercell) [6, 7]. For a chain of  $L$  sites labeled  $j = 0, 1, \dots, L-1$ , the Hamiltonian reads

$$\mathcal{H} = - \sum_{j=0}^{L-2} t_j (|j\rangle \langle j+1| + |j+1\rangle \langle j|), \quad (1)$$

where

$$t_j = \begin{cases} v & \text{if } j \text{ is even (intracell),} \\ w & \text{if } j \text{ is odd (intercell).} \end{cases} \quad (2)$$

The unit cell contains two sites—sublattice  $a$  (even sites) and sublattice  $b$  (odd sites). In the thermodynamic limit ( $L \rightarrow \infty$ ), the Bloch Hamiltonian in momentum space is  $\mathcal{H}(k) = \mathbf{d}(k) \cdot \boldsymbol{\sigma}$ , where  $\boldsymbol{\sigma} = (\sigma_x, \sigma_y)$  and the winding of  $\mathbf{d}(k)$  around the origin classifies the topological phases.

### B. Energy spectrum and topological transition

The bulk dispersion relation is

$$E(k) = \pm \sqrt{v^2 + w^2 + 2vw \cos k}, \quad (3)$$

yielding a spectral gap

$$\Delta = 2|w - v| \quad (4)$$

that closes at  $v = w$ , signaling the topological phase transition. For  $v < w$  the system is in the *topological phase* with winding number  $\nu = 1$ , while for  $v > w$  it is in the *trivial phase* with  $\nu = 0$  [5, 15].

Figure 1 shows the energy spectrum of a finite SSH chain ( $L = 10$ ) as a function of  $v/w$ . The gap closes linearly at the critical point  $v = w$ . For a chain with an odd number of sites ( $L = 11$ ), a zero-energy edge state appears in the topological phase, manifesting the bulk-boundary correspondence [16].

### C. Topological invariant

The topological classification of the SSH model is captured by the *winding number*

$$\nu = \frac{1}{2\pi} \oint dk \frac{d}{dk} \arg[d_x(k) + i d_y(k)], \quad (5)$$

where  $d_x(k) = v + w \cos k$  and  $d_y(k) = w \sin k$ . The winding number counts how many times the vector  $\mathbf{d}(k)$  encircles the origin as  $k$  traverses the Brillouin zone. For the SSH model:

- $v < w$ :  $\mathbf{d}(k)$  encircles the origin  $\Rightarrow \nu = 1$  (topological),
- $v > w$ :  $\mathbf{d}(k)$  does not encircle the origin  $\Rightarrow \nu = 0$  (trivial).

The winding number is quantized and can only change when the gap closes ( $v = w$ ), reflecting its topological nature. By the bulk-boundary correspondence, a chain with  $\nu = 1$  hosts one protected zero-energy state per boundary.

### D. Domain walls and protected states

A *domain wall* (DW) is created at the interface between two SSH domains with different dimerization patterns. Consider a chain of total length  $L$  composed of two domains meeting at site  $j_{\text{DW}}$ :

- **Domain 1** ( $j < j_{\text{DW}}$ ): standard SSH hopping pattern  $(v, w, v, w, \dots)$ ,
- **Domain 2** ( $j \geq j_{\text{DW}}$ ): the pattern restarts as  $(v, w, v, w, \dots)$ .

When  $j_{\text{DW}}$  is at an *odd* site, the bond at the interface involves two consecutive  $v$ -bonds (a  $vv$  defect), which constitutes a topological domain wall. This is because the dimerization pattern on each side of the interface is conjugate: if the left domain is in the topological phase, the right domain effectively presents the reversed pattern.

For a chain with  $N = 2$  domains (one domain wall), the system hosts three topologically protected states near  $E = 0$  (Fig. 2):

$$|\mathcal{L}\rangle \propto \sum_{n=0}^{\ell_1/2} (-r)^n |2n\rangle, \quad (6)$$

$$|\mathcal{S}\rangle \propto |j_{\text{DW}}\rangle + \sum_{n=1}^{\ell_1/2} (-r)^n |j_{\text{DW}} - 2n\rangle + \sum_{n=1}^{\ell_2/2} (-r)^n |j_{\text{DW}} + 2n\rangle, \quad (7)$$

$$|\mathcal{R}\rangle \propto \sum_{n=0}^{\ell_2/2} (-r)^n |L - 1 - 2n\rangle, \quad (8)$$

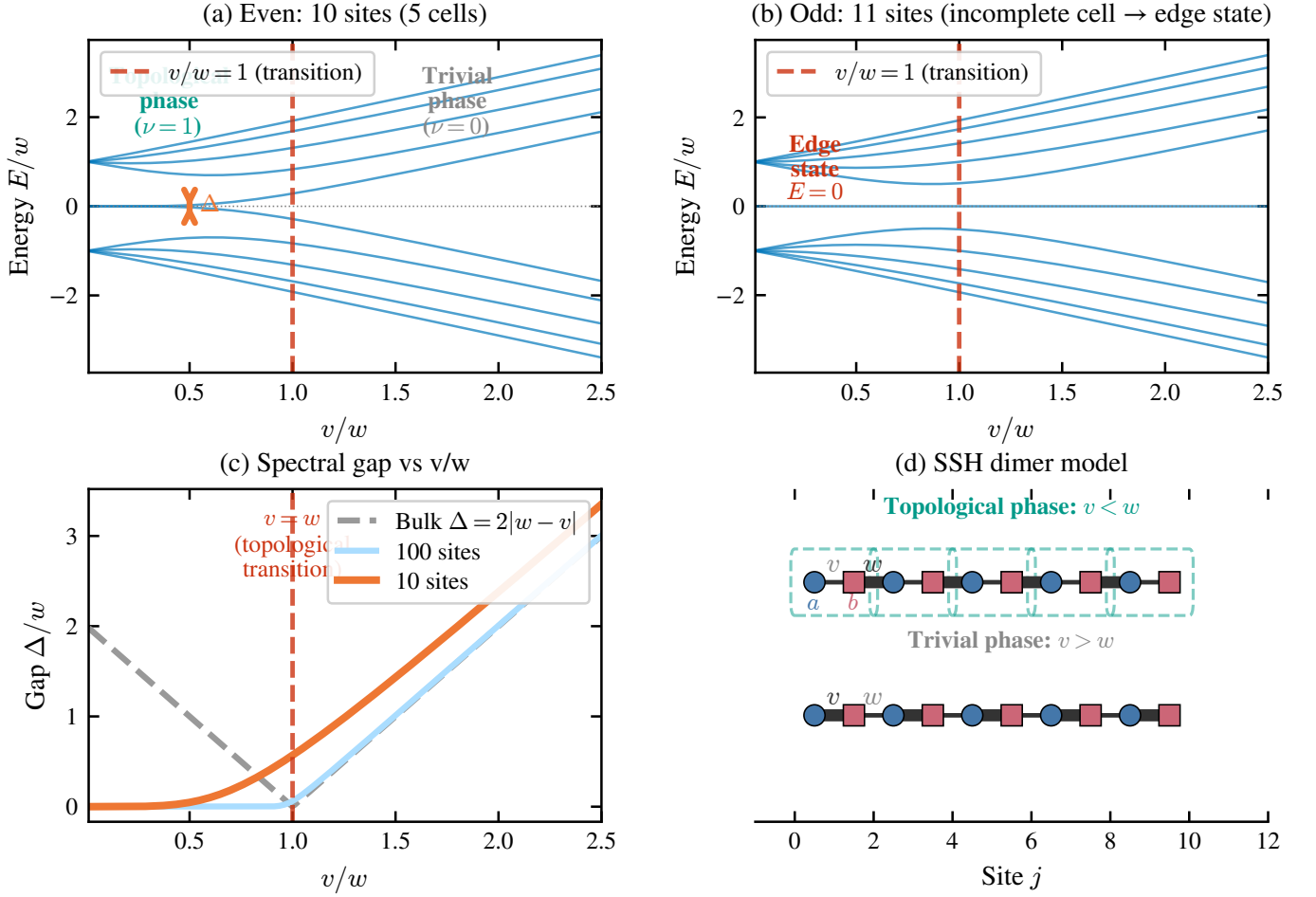


FIG. 1. Energy spectrum of the SSH dimer chain as a function of the hopping ratio  $v/w$ . **Left:** Even chain ( $L = 10$ ). The bulk gap  $\Delta = 2|w - v|$  closes at  $v = w$ . **Right:** Odd chain ( $L = 11$ ). A zero-energy state (red) appears in the topological phase ( $v/w < 1$ ), localized at the boundary.

where  $r = v/w$  is the decay ratio, and  $\ell_1, \ell_2$  are the lengths (number of bonds) of the left and right domains, respectively. Each state resides on a single sublattice and decays exponentially away from its localization center with characteristic length  $\xi = -1/\ln r$ .

### E. Effective low-energy Hamiltonian

The three protected states span an effective low-energy subspace described by

$$\mathcal{H}_{\text{eff}} = J_{\mathcal{L}\mathcal{S}} |\mathcal{S}\rangle\langle\mathcal{L}| + J_{\mathcal{S}\mathcal{R}} |\mathcal{R}\rangle\langle\mathcal{S}| + \text{H.c.}, \quad (9)$$

which has the structure of a three-site tight-binding chain. The effective couplings  $J_{\mathcal{L}\mathcal{S}}$  and  $J_{\mathcal{S}\mathcal{R}}$  depend on the overlap of adjacent protected states through the SSH bulk. From the analytical expressions [Eqs. (6)–(8)], these couplings scale as [9]

$$J \sim v \cdot \mathcal{M}_i \cdot \mathcal{M}_j \cdot \left(\frac{v}{w}\right)^{\ell/2}, \quad (10)$$

where  $\ell$  is the domain length separating the two states, and the normalization constants are

$$\mathcal{M}_{\mathcal{L}} = \mathcal{M}_{\mathcal{R}} = \sqrt{\frac{w^2}{v^2} - 1}, \quad \mathcal{M}_{\mathcal{S}} = \sqrt{\frac{w^2 - v^2}{w^2 + v^2}}. \quad (11)$$

*Crucially*, the exponential dependence of  $J$  on  $\ell$  implies that even a small asymmetry  $\ell_1 \neq \ell_2$  leads to vastly different effective couplings:

$$\frac{J_{\mathcal{L}\mathcal{S}}}{J_{\mathcal{S}\mathcal{R}}} \sim \left(\frac{w}{v}\right)^{(\ell_2 - \ell_1)/2}. \quad (12)$$

For our parameters  $w = 2v$ , a difference  $|\ell_1 - \ell_2| = 2$  already produces a coupling ratio of 4, while  $|\ell_1 - \ell_2| = 8$  yields a ratio of  $2^4 = 16$ .

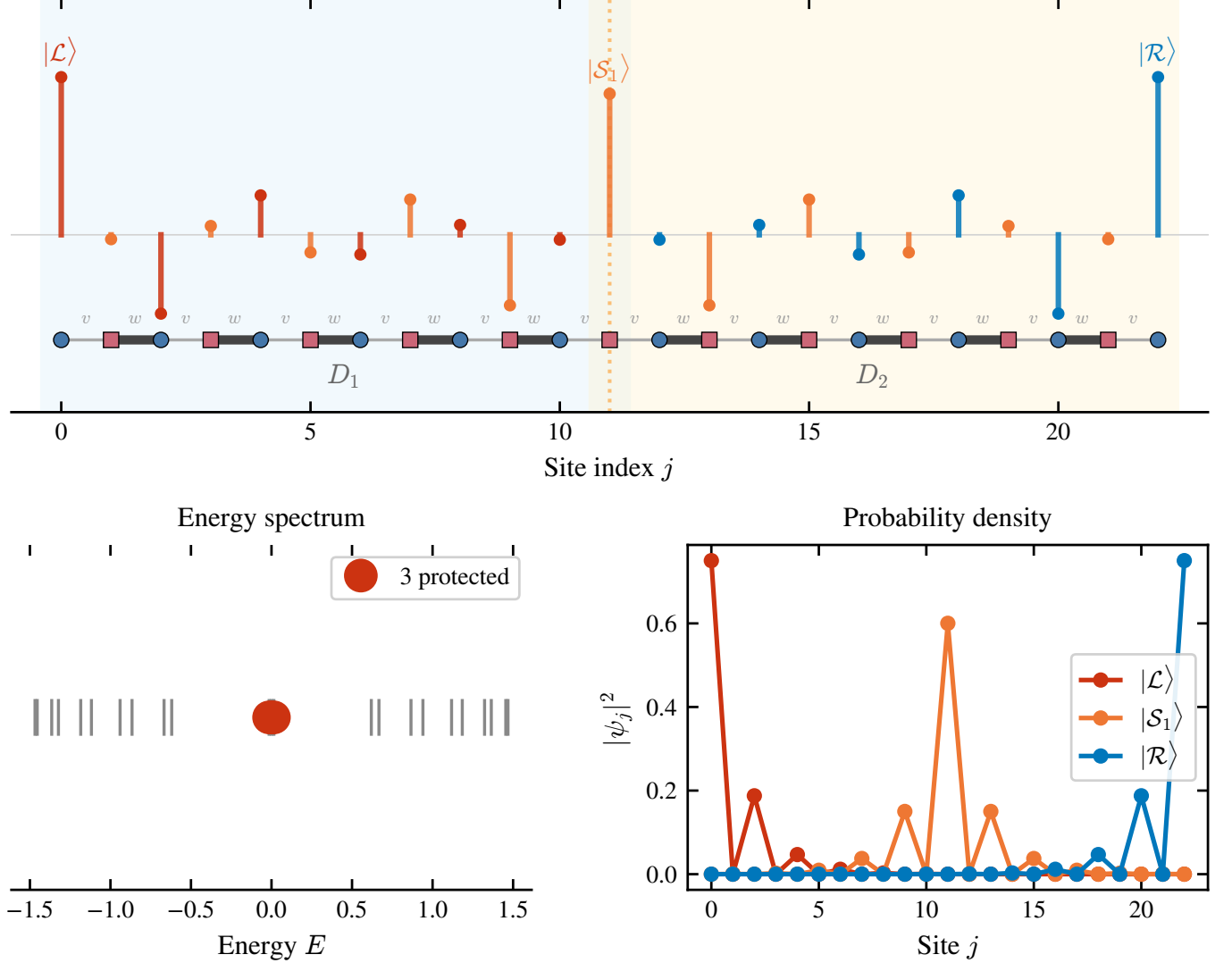
SSH chain with 1 domain wall:  $N = 2, \ell = 10, L = 23, w = 2v = 1$ 

FIG. 2. Topologically protected states in a two-domain SSH chain ( $L = 11, N = 2, \ell = 4, v = 0.5, w = 1.0$ ). The three states  $|\mathcal{L}\rangle$ ,  $|\mathcal{S}\rangle$ , and  $|\mathcal{R}\rangle$  are localized at the left edge, the domain wall, and the right edge, respectively. Sublattice polarization (alternating signs) is clearly visible.

### III. TRANSFER PROTOCOL

#### A. Adiabatic preparation and readout

The transfer protocol proposed by J. Zurita, C. E. Creffield, and G. Platero [9] consists of three phases:

1. **Preparation** ( $0 \leq t < t_{\text{prep}}$ ): The intracell hopping is ramped from  $v = 0$  (fully dimerized) to  $v = v_{\text{tr}}$  following a smooth pulse shape. This adiabatically maps the initial localized state  $|0\rangle$  onto the protected boundary state  $|\mathcal{L}\rangle$ .
2. **Transfer** ( $t_{\text{prep}} \leq t < t_{\text{tr}} - t_{\text{prep}}$ ): The hopping is held constant at  $v = v_{\text{tr}}$ . The effective couplings

$J_{\mathcal{L}\mathcal{S}}$ ,  $J_{\mathcal{S}\mathcal{R}}$  drive Rabi-like oscillations in the protected subspace, transferring population from  $|\mathcal{L}\rangle$  to  $|\mathcal{R}\rangle$  via  $|\mathcal{S}\rangle$ .

3. **Readout** ( $t_{\text{tr}} - t_{\text{prep}} \leq t \leq t_{\text{tr}}$ ): The hopping is ramped back to  $v = 0$ , adiabatically mapping  $|\mathcal{R}\rangle$  back onto the localized state  $|L-1\rangle$ .

The standard pulse shape (Eq. (11) of Ref. [9]) is

$$v(t) = \begin{cases} v_{\text{tr}} \sin^2(\Omega t) & 0 \leq t < t_{\text{prep}}, \\ v_{\text{tr}} & t_{\text{prep}} \leq t < t_{\text{tr}} - t_{\text{prep}}, \\ v_{\text{tr}} \sin^2(\Omega(t - t_{\text{tr}})) & t_{\text{tr}} - t_{\text{prep}} \leq t \leq t_{\text{tr}}, \end{cases} \quad (13)$$

with  $\Omega = \pi/(2t_{\text{prep}})$ . The  $\sin^2$  ramp is chosen for its smoothness: both  $v(t)$  and  $\dot{v}(t)$  vanish at  $t = 0$  and  $t =$

$t_{\text{prep}}$ , minimizing diabatic transitions to bulk states. The adiabatic condition requires  $t_{\text{prep}} \gg \tau = 2/\Delta$ , where  $\Delta$  is the instantaneous spectral gap.

The transfer fidelity is defined as the occupation probability of the target site at the end of the protocol:

$$f = |\langle L-1 | \psi(t_{\text{tr}}) \rangle|^2. \quad (14)$$

### B. Transfer time

For a symmetric two-domain chain ( $\ell_1 = \ell_2 = \ell$ ), the effective Hamiltonian (9) has equal couplings  $J_{\mathcal{L}\mathcal{S}} = J_{\mathcal{S}\mathcal{R}} \equiv J$ . The time for complete population transfer from  $|\mathcal{L}\rangle$  to  $|\mathcal{R}\rangle$  via the intermediate state  $|\mathcal{S}\rangle$  in a uniform three-site chain is

$$t_{\text{tr}}^{\text{eff}} = \frac{\pi}{\sqrt{2}J}, \quad (15)$$

which, using Eq. (10), scales exponentially with the domain length:

$$t_{\text{tr}} \sim \frac{\pi(w^2 + v^2)}{2(w^2 - v^2)} \left(\frac{w}{v}\right)^{\ell/2+1}. \quad (16)$$

### C. Numerical results: reproducing Ref. [9]

We simulate the full time-dependent Schrödinger equation for a chain of  $L = 11$  sites ( $N = 2$  domains,  $\ell = 4$ ) with  $w = 1.0$ ,  $v_{\text{tr}} = 0.5$ , and  $t_{\text{prep}} = 15$ . The time evolution is performed via exact exponentiation of the Hamiltonian at each time step  $\Delta t = 0.5$ .

Figure 3 shows the transfer dynamics for  $t_{\text{tr}} = 45.0$ . The initial state  $|0\rangle$  is clearly seen to evolve through the three protected states: population flows from the left edge to the domain wall and then to the right edge, achieving a final fidelity  $f = 0.999$ .

Figure 4 presents the fidelity as a function of  $t_{\text{tr}}$ , showing Rabi-like oscillations with the first maximum at  $t_{\text{tr}} \approx 45$ , in excellent agreement with the theoretical prediction  $t_{\text{tr}} \approx 45.6$  from Eq. (16) and the value reported in Ref. [9]. The peak fidelity  $f = 0.999$  exceeds the threshold  $f_0 = 0.995$  commonly cited for fault-tolerant quantum error correction [17].

## IV. ASYMMETRIC DOMAIN WALL PLACEMENT

### A. Motivation

In practice, fabrication imperfections may shift the domain wall away from the ideal center of the chain. Furthermore, for chains of certain lengths, it is geometrically impossible to place the domain wall exactly at the center. The domain wall position  $j_{\text{DW}}$  must be at an *odd-index* site for the  $vv$ -bond defect to occur; thus, only chains

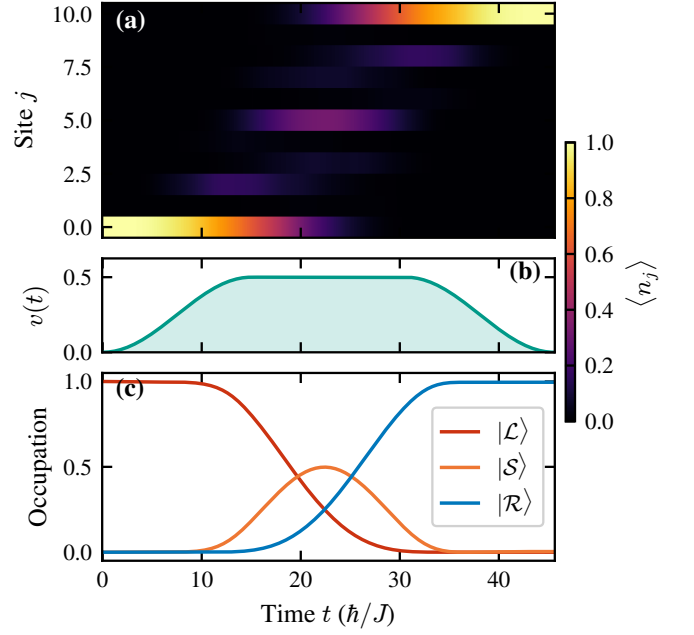


FIG. 3. Transfer protocol dynamics for the symmetric two-domain SSH chain ( $L = 11$ ,  $\ell = 4$ ,  $v_{\text{tr}} = 0.5$ ,  $w = 1.0$ ,  $t_{\text{prep}} = 15$ ,  $t_{\text{tr}} = 45$ ). Population is transferred from site 0 to site 10 via the topologically protected subspace, with final fidelity  $f = 0.999$ .

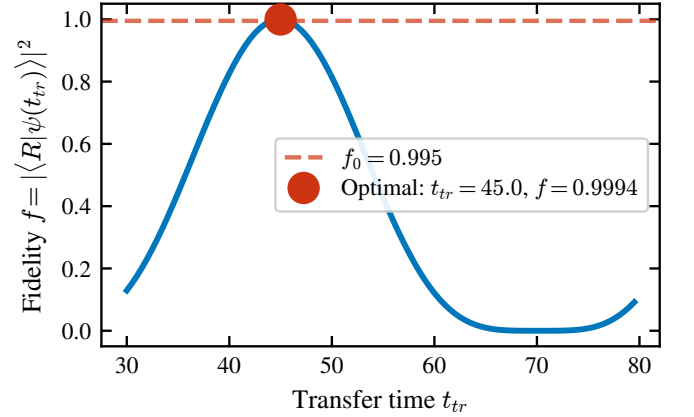


FIG. 4. Transfer fidelity as a function of the total protocol time  $t_{\text{tr}}$  for the symmetric chain ( $L = 11$ ,  $\ell = 4$ ). The first maximum occurs at  $t_{\text{tr}} \approx 45$  with  $f = 0.999$ , consistent with Ref. [9]. Subsequent oscillations arise from the Rabi dynamics of the effective three-site chain.

with  $L = 4k + 3$  (i.e.,  $L = 11, 15, 19, 23, \dots$ ) admit a perfectly centered domain wall with  $\ell_1 = \ell_2$ .

We study a chain of  $L = 21$  sites (10 complete unit cells plus one boundary site), varying the domain wall position among three representative odd sites:

- $j_{\text{DW}} = 11$  (quasi-center):  $\ell_1 = 11$ ,  $\ell_2 = 9$ ,
- $j_{\text{DW}} = 7$  (one-third):  $\ell_1 = 7$ ,  $\ell_2 = 13$ ,
- $j_{\text{DW}} = 5$  (one-quarter):  $\ell_1 = 5$ ,  $\ell_2 = 15$ .

## B. Protected states in asymmetric configurations

Regardless of the domain wall position, the system always hosts three protected states near  $E = 0$ , given by Eqs. (6)–(8) with  $\ell_1 \neq \ell_2$ . We verify this numerically by diagonalizing the full Hamiltonian and comparing the three eigenstates closest to  $E = 0$  with the analytical predictions. Table I summarizes the results.

Figure 5 illustrates how the spatial profiles of the protected states change with the domain wall position. As the wall moves toward the left edge,  $|\mathcal{L}\rangle$  becomes increasingly compressed into a shorter domain (fewer sites available for its exponential tail), while  $|\mathcal{R}\rangle$  extends over a longer domain. The state  $|\mathcal{S}\rangle$  develops an asymmetric envelope, decaying faster into the shorter domain.

## C. Effective couplings and the asymmetry mechanism

The effective couplings  $J_{\mathcal{LS}}$  and  $J_{\mathcal{SR}}$  are computed numerically by projecting the full Hamiltonian onto the protected subspace. The results (Fig. 6) confirm the exponential scaling predicted by Eq. (12):

The coupling asymmetry has a direct and dramatic consequence for the transfer fidelity. In a three-site chain with unequal hoppings  $J_1 \neq J_2$ , the maximum achievable population transfer from site 1 to site 3 is bounded by [2]

$$f_{\max} = \frac{4J_1^2 J_2^2}{(J_1^2 + J_2^2)^2}, \quad (17)$$

which equals unity only when  $J_1 = J_2$  and decreases rapidly with the coupling ratio. For our quasi-centered case ( $J_{\mathcal{LS}}/J_{\mathcal{SR}} = 0.5$ ):  $f_{\max} = 4 \times 0.25/(1.25)^2 = 16/25 = 0.64$ .

## D. Transfer fidelity in asymmetric chains

Figure 7 shows the transfer dynamics for the three asymmetric configurations, using the standard  $\sin^2$  pulse with  $t_{\text{prep}} = 15$ . Table III reports the optimal fidelities obtained from a scan over  $t_{\text{tr}} \in [30, 600]$ .

The quasi-centered case ( $j_{\text{DW}} = 11$ ) achieves  $f = 0.637$ , in quantitative agreement with the analytical bound  $f_{\max} = 0.64$  from Eq. (17). The small discrepancy arises from residual diabatic excitations during the

ramp phases. For  $j_{\text{DW}} = 7$  and 5, the fidelity drops to  $f = 0.061$  and  $f = 0.004$ , respectively—effectively rendering the transfer useless.

Figure 8 presents a direct comparison of the fidelity-vs- $t_{\text{tr}}$  curves for all three wall positions. The oscillation frequencies and amplitudes differ markedly, reflecting the distinct effective coupling landscapes.

## E. Systematic study: fidelity vs. wall position

To map out the full dependence of the transfer fidelity on domain wall position, we perform systematic scans for chains of length  $L = 11$  and  $L = 15$ , placing the domain wall at all allowed (odd) sites. Figure 9 shows that the fidelity profile is perfectly symmetric about the chain center (reflecting the equivalence of swapping the two domains) and sharply peaked:

- $L = 11$ : peak fidelity  $f = 0.96$  at  $j_{\text{DW}} = 5$  (center), dropping to  $f = 0.20$  at  $j_{\text{DW}} = 3$  and  $f = 0.004$  at  $j_{\text{DW}} = 1$ .
- $L = 15$ : peak fidelity  $f = 0.994$  at  $j_{\text{DW}} = 7$  (center), dropping to  $f = 0.22$  at  $j_{\text{DW}} = 5$  and  $f = 0.014$  at  $j_{\text{DW}} = 3$ .

TABLE I. Protected states for the asymmetric domain wall configurations in an  $L = 21$  chain ( $v = 0.5$ ,  $w = 1.0$ ). The subspace overlap is computed as the singular values of the overlap matrix between analytical and numerical eigenstates.

| $j_{\text{DW}}$ | $\ell_1 + \ell_2$ | $E_{\pm}$   | Min. overlap | $J_{\mathcal{LS}}/J_{\mathcal{SR}}$ |
|-----------------|-------------------|-------------|--------------|-------------------------------------|
| 11              | 11 + 9            | $\pm 0.024$ | 0.9998       | 0.50                                |
| 7               | 7 + 13            | $\pm 0.043$ | 0.9994       | 8.0                                 |
| 5               | 5 + 15            | $\pm 0.086$ | 0.9978       | 32.0                                |

TABLE II. Effective couplings for the three domain wall positions ( $L = 21$ ,  $v = 0.5$ ,  $w = 1.0$ ).

| $j_{\text{DW}}$ | $J_{\mathcal{LS}}$ | $J_{\mathcal{SR}}$ | Ratio | Predicted ratio  |
|-----------------|--------------------|--------------------|-------|------------------|
| 11              | 0.0148             | 0.0296             | 0.50  | $(v/w)^1 = 0.50$ |
| 7               | 0.0593             | 0.0074             | 8.0   | $(w/v)^3 = 8.0$  |
| 5               | 0.1186             | 0.0037             | 32.0  | $(w/v)^5 = 32.0$ |

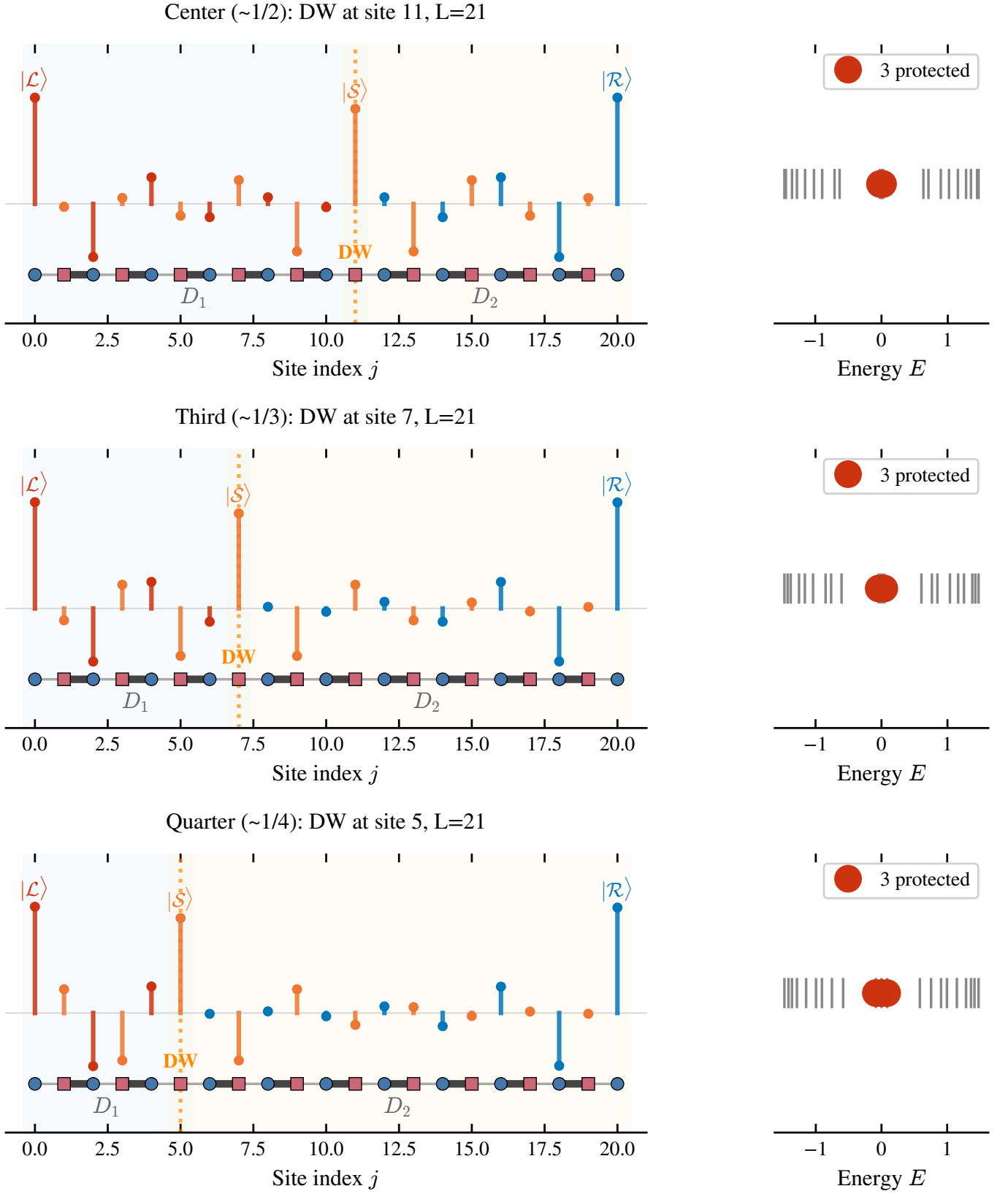


FIG. 5. Spatial profiles of the three protected states for different domain wall positions in an  $L = 21$  chain. Top to bottom:  $j_{\text{DW}} = 11$  (quasi-center),  $j_{\text{DW}} = 7$  (one-third),  $j_{\text{DW}} = 5$  (one-quarter). The exponential localization and sublattice polarization are preserved in all cases.

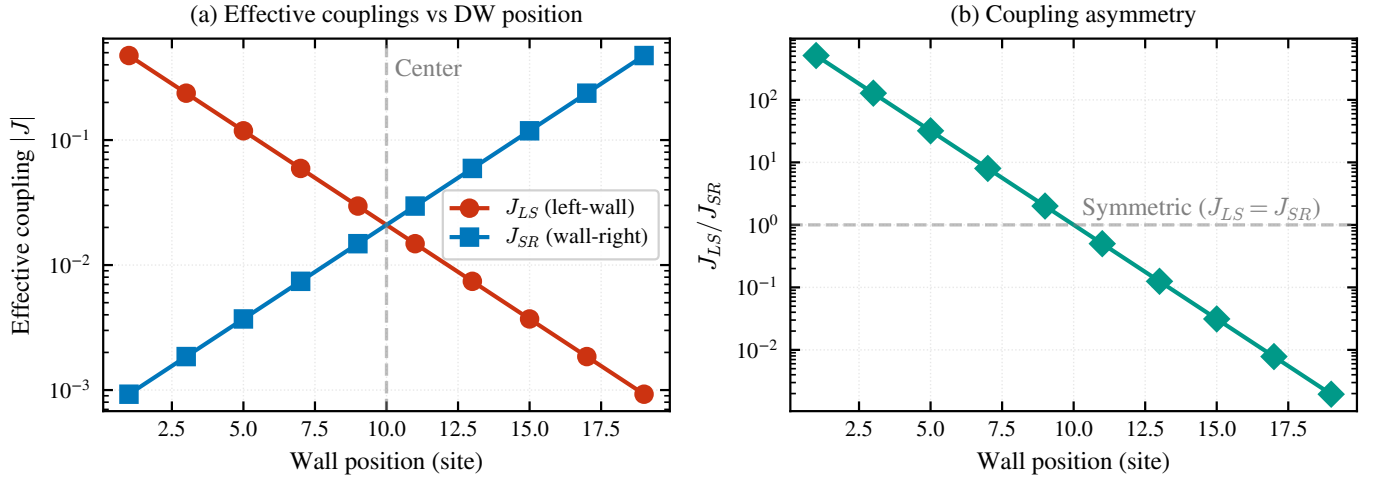


FIG. 6. Effective couplings  $J_{LS}$  (left-soliton) and  $J_{SR}$  (soliton-right) as a function of the domain wall position for  $L = 21$ . The coupling asymmetry grows exponentially with the displacement from the chain center, consistent with Eq. (12).



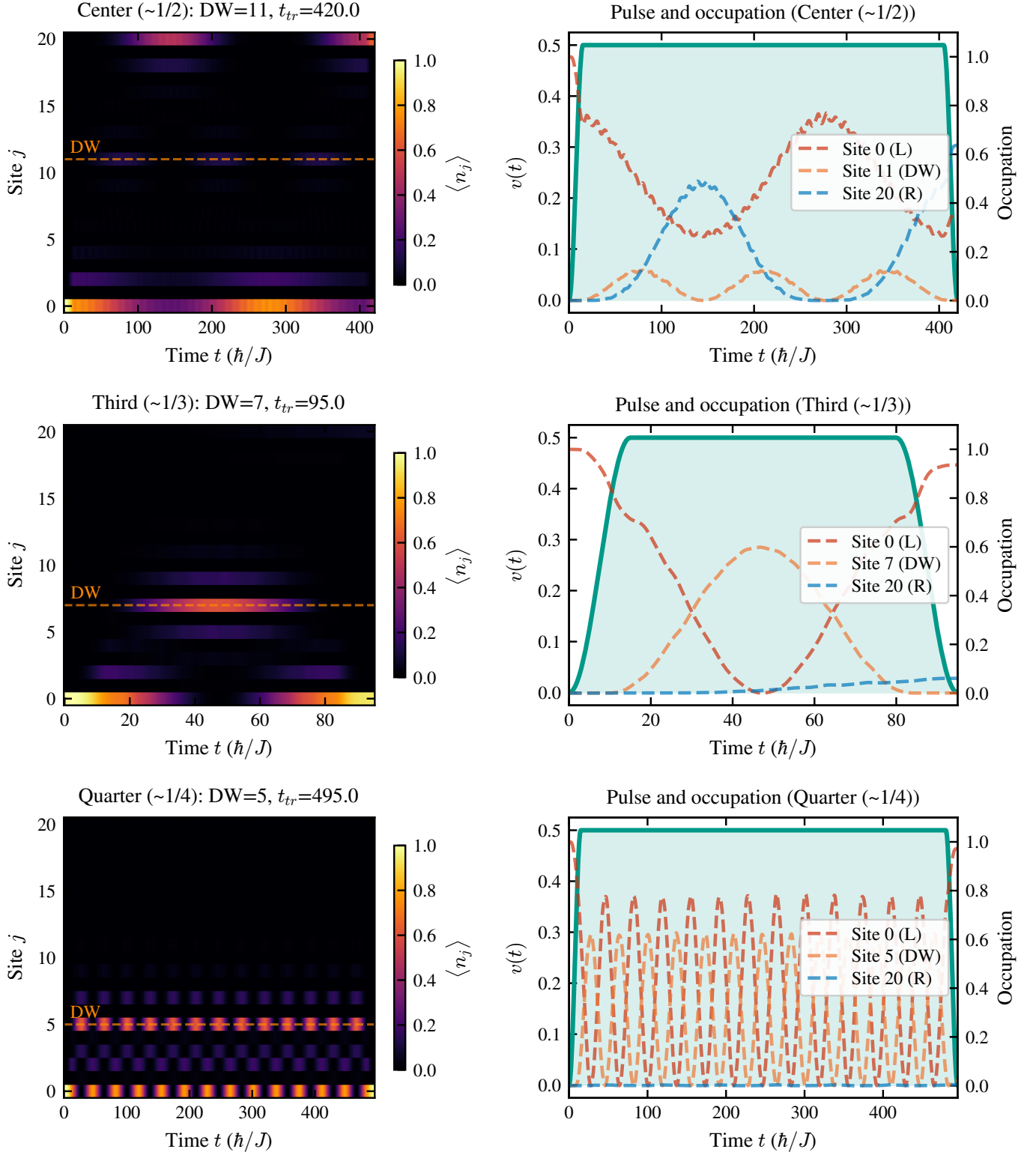


FIG. 7. Transfer dynamics for asymmetric domain wall configurations ( $L = 21$ ). The fidelity degrades dramatically as the domain wall moves away from the chain center.

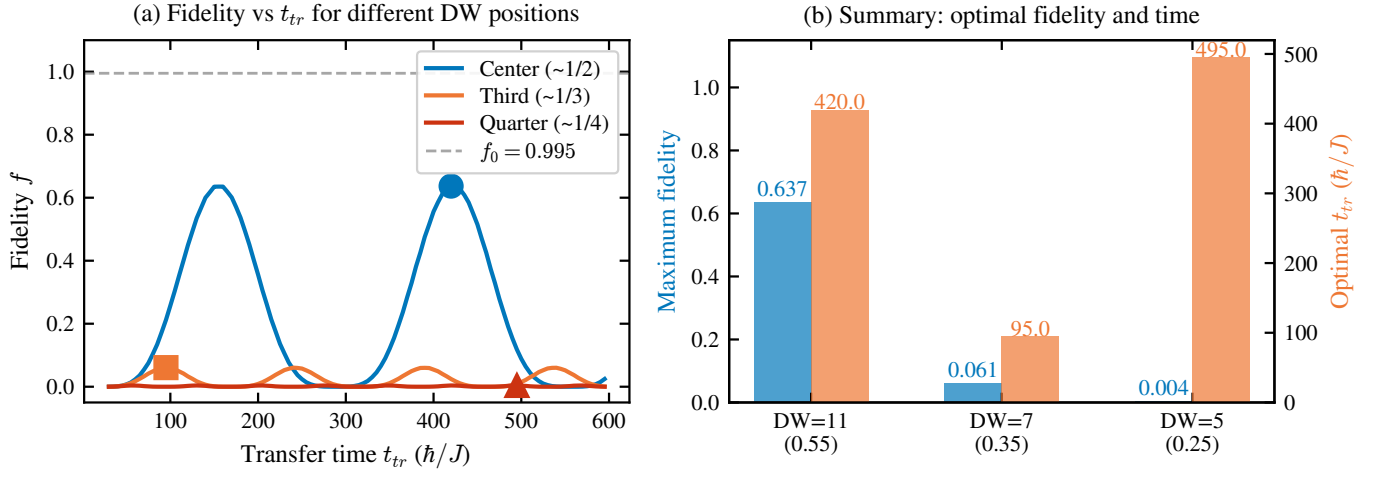


FIG. 8. Transfer fidelity as a function of  $t_{tr}$  for the three domain wall positions in the  $L = 21$  chain. The first fidelity maximum (marked by dots) decreases exponentially with the displacement from the center.

TABLE III. Optimal transfer fidelity for asymmetric domain wall positions ( $L = 21$ , standard  $\sin^2$  pulse).

| $j_{\text{DW}}$   | $\ell_1 + \ell_2$ | $t_{\text{tr}}^{\text{opt}}$ | $f_{\text{max}}$ |
|-------------------|-------------------|------------------------------|------------------|
| 11 (quasi-center) | 11 + 9            | 420                          | 0.637            |
| 7 (one-third)     | 7 + 13            | 95                           | 0.061            |
| 5 (one-quarter)   | 5 + 15            | 495                          | 0.004            |

The symmetric profiles confirm that the transfer fidelity depends only on the *difference*  $|\ell_1 - \ell_2|$  between domain lengths, not on their individual values. The exponential sensitivity to this difference—inherited from Eq. (12)—imposes stringent requirements on the precision of the domain wall placement in experimental implementations.

## V. SHORTCUT-TO-ADIABATICITY PROTOCOLS

### A. Motivation and theoretical framework

The preparation and readout phases of the transfer protocol involve adiabatic ramps of the intracell hopping  $v(t)$ . When the ramp time  $t_{\text{prep}}$  is not sufficiently long compared to the inverse gap  $\tau = 2/\Delta$ , diabatic excitations to bulk states reduce the fidelity. Shortcut-to-adiabaticity (STA) techniques [10–12, 18] aim to suppress these excitations by modifying the pulse shape, potentially allowing faster operation without sacrificing fidelity.

The exact counter-diabatic Hamiltonian [11]

$$H_{\text{CD}}(t) = i\hbar \sum_n (|\dot{n}(t)\rangle\langle n(t)| - \langle n(t)|\dot{n}(t)\rangle |n(t)\rangle\langle n(t)|) \quad (18)$$

cancels all diabatic transitions exactly but requires non-local interactions that are impractical to implement. Instead, we explore *approximate* STA strategies through pulse shape optimization.

### B. Pulse profiles

We compare four pulse profiles for the ramp phases (Fig. 10):

1. **Standard**  $\sin^2$  [Eq. (13)]: The reference protocol from Ref. [9].
2. **STA- $\alpha$** : A modified ramp  $v(t) = v_{\text{tr}} \sin^{2\alpha}(\Omega t)$  with  $\alpha > 1$ . The higher exponent produces a steeper ramp that spends less time in the intermediate- $v$  region where the gap is smaller. We test  $\alpha = 2$  and  $\alpha = 3$ .
3. **STA-global**  $\sin^2$ : A single smooth profile over the entire protocol:

$$v(t) = v_{\text{tr}} \sin^2\left(\frac{\pi t}{t_{\text{tr}}}\right), \quad (19)$$

which has no plateau phase and is infinitely differentiable everywhere.

4. **Linear ramp**:  $v(t) = v_{\text{tr}} \min(t/t_{\text{prep}}, 1, (t_{\text{tr}} - t)/t_{\text{prep}})$ , serving as a baseline with discontinuous  $\dot{v}$  that generates stronger diabatic excitations.

### C. Results: symmetric chain

Table IV and Fig. 10 summarize the performance of each pulse profile on the symmetric chain ( $L = 11$ ,  $\ell = 4$ ).

TABLE IV. Pulse comparison for the symmetric chain ( $L = 11$ ,  $\ell = 4$ ,  $v_{\text{tr}} = 0.5$ ,  $w = 1.0$ ).

| Pulse               | $t_{\text{prep}}$ | $t_{\text{tr}}^{\text{opt}}$ | $f_{\text{max}}$ |
|---------------------|-------------------|------------------------------|------------------|
| Standard $\sin^2$   | 15                | 45                           | 0.9994           |
| Standard $\sin^2$   | 10                | 89                           | 0.9928           |
| Standard $\sin^2$   | 5                 | 33                           | 0.9290           |
| STA $\alpha = 2$    | 10                | 41                           | 0.9706           |
| STA $\alpha = 3$    | 10                | 43                           | 0.9398           |
| STA-global $\sin^2$ | —                 | 75                           | 0.99995          |
| Linear              | 15                | 97                           | 0.9924           |

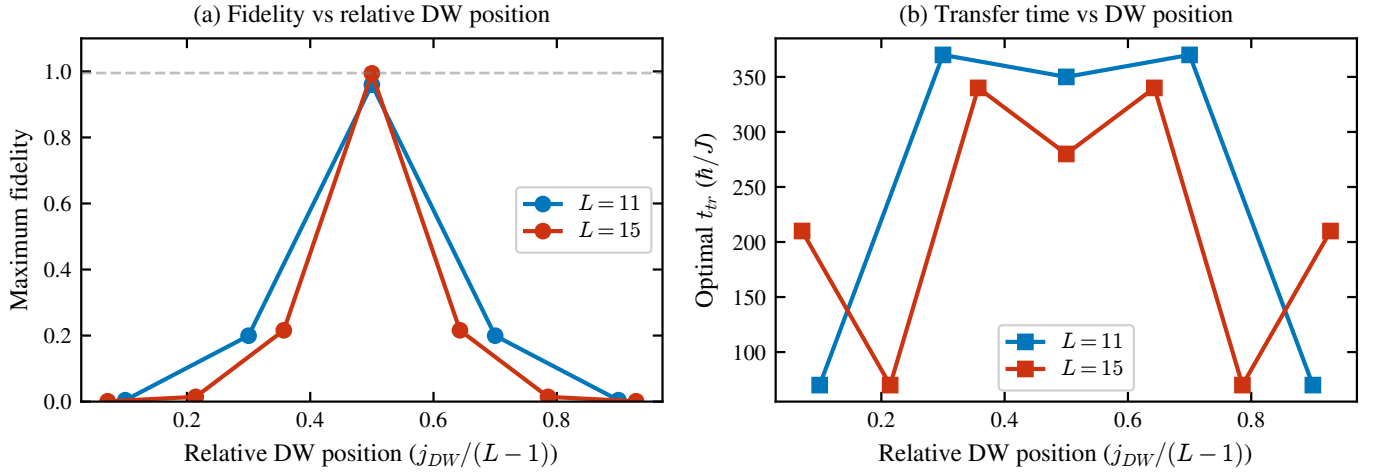


FIG. 9. Maximum transfer fidelity (left panel) and optimal transfer time  $t_{tr}$  (right panel) as a function of the normalized domain wall position  $j_{DW}/(L-1)$ , for chains of length  $L=11$  (blue) and  $L=15$  (orange). Both quantities peak sharply at the chain center ( $j_{DW}/(L-1) = 0.5$ ): the fidelity reaches its maximum and the optimal transfer time its minimum. Only odd-index sites produce valid domain walls.

The key findings are:

- The standard  $\sin^2$  pulse with  $t_{\text{prep}} = 15$  provides the optimal balance between speed and fidelity, achieving  $f = 0.9994$  at  $t_{\text{tr}} = 45$ —reproducing the result of Ref. [9].
- The STA-global  $\sin^2$  pulse achieves the highest absolute fidelity ( $f = 0.99995$ ) but at the cost of a 67% longer protocol time ( $t_{\text{tr}} = 75$  vs. 45).
- The STA- $\alpha$  pulses with reduced  $t_{\text{prep}} = 10$  do *not* improve upon the standard protocol. The steeper ramp increases diabatic excitations rather than suppressing them, as the system spends insufficient time adjusting to the changing Hamiltonian.
- The linear ramp performs worst in terms of speed ( $t_{\text{tr}} = 97$ ) due to the discontinuities in  $\dot{v}$  at the ramp endpoints, which generate diabatic excitations.

#### D. Preparation time dependence

Figure 11 shows the dependence of the optimal fidelity on the preparation time  $t_{\text{prep}}$  for the standard and STA- $\alpha$  pulses. The fidelity follows a characteristic sigmoid: for  $t_{\text{prep}} \ll \tau$  the ramp is too fast and diabatic transitions dominate, while for  $t_{\text{prep}} \gtrsim \tau$  the fidelity saturates near its maximum. The crossover occurs at  $t_{\text{prep}} \sim 2/\Delta_{\text{min}}$ , where  $\Delta_{\text{min}}$  is the minimum spectral gap encountered during the ramp.

#### E. Results: asymmetric chains with STA

A natural question is whether STA protocols can compensate for the fidelity degradation caused by asymmetric domain wall placement. We test the standard, STA- $\alpha = 2$ , and STA-global  $\sin^2$  pulses on the three asymmetric configurations ( $L = 21$ ,  $j_{\text{DW}} = 11, 7, 5$ ).

The results (Table V and Fig. 12) are unambiguous: *no pulse protocol significantly improves the fidelity in asymmetric configurations*. This confirms that the limitation is not due to diabatic excitations during the ramp phases—which STA protocols are designed to address—but rather stems from the fundamental coupling asymmetry  $J_{\mathcal{LS}} \neq J_{\mathcal{SR}}$  in the transfer phase. The pulse shape

TABLE V. Pulse comparison for asymmetric configurations ( $L = 21$ ). No pulse protocol significantly improves the fidelity, confirming that the limitation is topological.

| $j_{\text{DW}}$   | Standard | STA $\alpha = 2$ | STA-global |
|-------------------|----------|------------------|------------|
| 11 (quasi-center) | 0.637    | 0.610            | 0.609      |
| 7 (one-third)     | 0.061    | 0.060            | 0.047      |
| 5 (one-quarter)   | 0.004    | 0.005            | 0.003      |

affects only the preparation and readout fidelity; **the transfer dynamics are governed entirely by the topological properties of the chain.**

### VI. HYBRID ANALOG-DIGITAL COUNTERDIABATIC PROTOCOL

In this section, we present a detailed analysis of the alternative transfer paradigm proposed by S. V. Romero, X. Chen, G. Platero, and Y. Ban [13]. Rather than exploiting topological domain walls as in the Zurita protocol (Sec. III), this approach achieves nonadiabatic edge-state transfer in a *uniform* SSH chain by supplementing the Hamiltonian with approximate counterdiabatic (CD) interactions. The CD terms are derived analytically from adiabatic gauge potentials and can be further optimized digitally via variational quantum circuits, hence the denomination “hybrid analog-digital.”

#### A. SSH chain with an odd number of sites

The starting point is a standard SSH chain with  $2N - 1$  sites (an odd number), with  $N$  sites on sublattice A (odd indices) and  $N - 1$  on sublattice B (even indices). The Hamiltonian is [13]

$$H_0(t) = t_2(t) \sum_{j=1}^{N-1} c_{2j}^\dagger c_{2j-1} + t_1(t) \sum_{j=1}^{N-1} c_{2j+1}^\dagger c_{2j} + \text{H.c.}, \quad (20)$$

where  $c_i^\dagger$  ( $c_i$ ) creates (annihilates) a fermion at site  $i$ ,  $t_2(t)$  is the intra-cell hopping, and  $t_1(t)$  is the inter-cell hopping. Due to chiral symmetry, the spectrum consists of  $2N - 2$  nonzero-energy levels arranged symmetrically about  $E = 0$ , plus one exact zero-energy eigenstate [5]:

$$|\Phi_0(t)\rangle = \frac{1}{\mathcal{N}} \sum_{i=1}^N \left[ -\frac{t_2(t)}{t_1(t)} \right]^{i-1} |A_i\rangle, \quad (21)$$

where  $\mathcal{N}^2 = \sum_{i=0}^{N-1} [t_2(t)/t_1(t)]^{2i}$  is the normalization factor. Crucially, this zero-energy state has nonzero components only on sublattice A sites (the  $b_i = 0$  property), a consequence of chiral symmetry that will be exploited in the simplification of the CD terms.

The transfer mechanism is conceptually simple: when  $t_1(0)/t_2(0) \gg 1$ , the zero-energy state is localized at the left edge (site  $A_1$ ). When  $t_1(T)/t_2(T) \ll 1$  at the final time  $t = T$ , the state migrates to the right edge (site  $A_N$ ). An adiabatic modulation of the hopping parameters achieves this transfer, as in the original Thouless pumping [19, 20]. A standard parametrization is [13]

$$t_{1,2}(t) = t_0(1 \pm \cos \Omega t), \quad (22)$$

where  $\Omega$  sets the speed of the protocol and the adiabatic regime corresponds to  $\Omega = 0.01 t_0$  (transfer time  $T =$

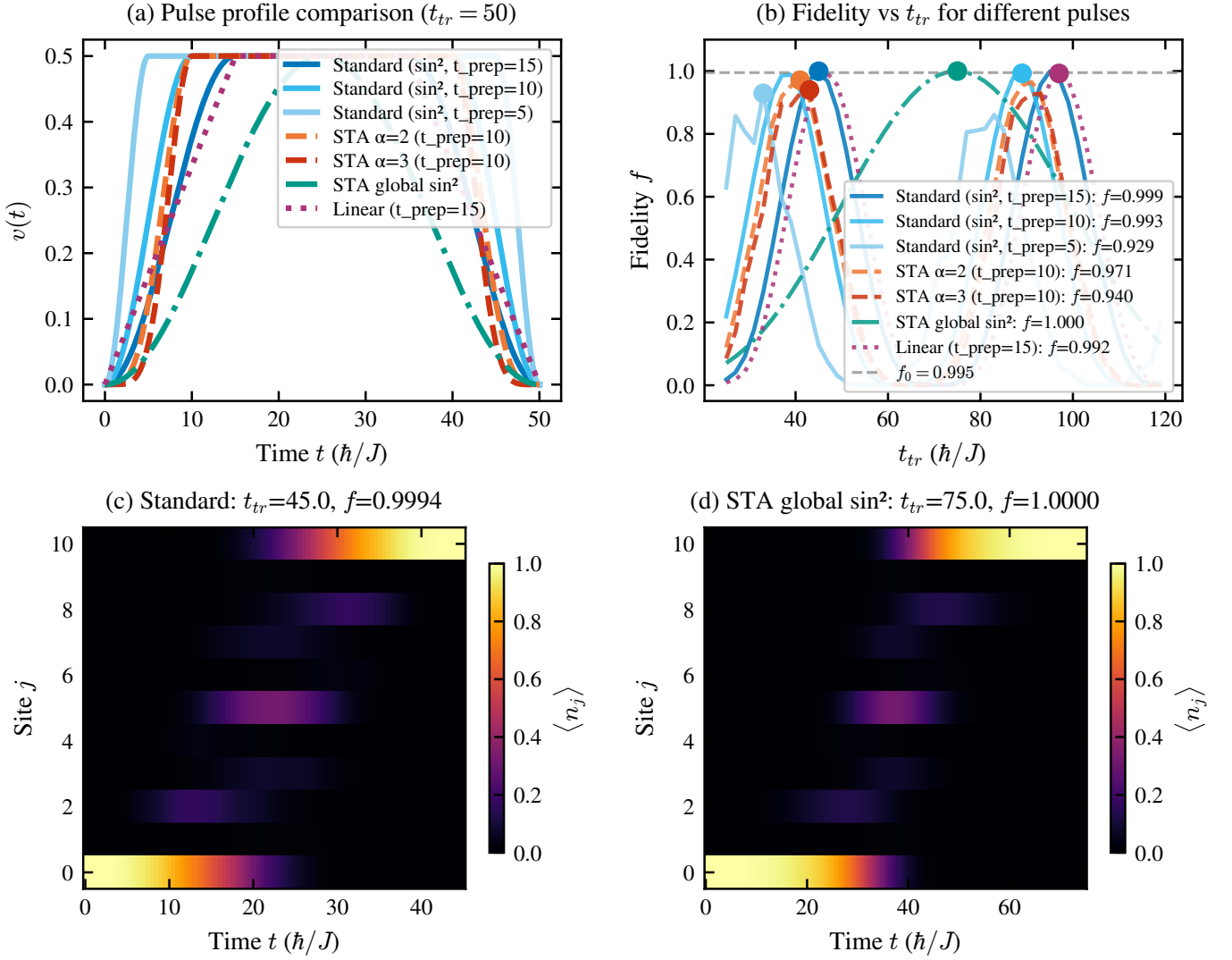


FIG. 10. Comparison of transfer fidelity vs.  $t_{tr}$  for different pulse profiles in the symmetric chain ( $L = 11$ ,  $\ell = 4$ ). The standard  $\sin^2$  pulse with  $t_{\text{prep}} = 15$  achieves the best time-fidelity trade-off ( $f = 0.9994$  at  $t_{tr} = 45$ ). The STA-global  $\sin^2$  achieves the highest absolute fidelity ( $f = 0.99995$ ) but requires  $t_{tr} = 75$ .

$\pi/\Omega = 100\pi/t_0$ ). The transfer fidelity is

$$F(t) = |\langle \Phi_0(t) | \Psi(t) \rangle|^2, \quad (23)$$

measuring how closely the evolved state  $|\Psi(t)\rangle$  follows the instantaneous zero-energy eigenstate at all times during the protocol.

It is important to note two key differences with the Zurita protocol (Sec. III A): (i) the chain contains no domain wall—it is a uniform SSH chain with a single dimerization pattern; and (ii) the transfer operates by dynamically sweeping the hopping ratio  $t_1(t)/t_2(t)$  through the entire topological phase diagram, rather than by populating the protected subspace at a fixed hopping ratio. The adiabatic condition  $T \gg \pi/t_0$  ensures that the state tracks the instantaneous eigenstate, but this requires very long protocol times, making the system vulnerable to decoherence.

## B. Counterdiabatic driving from adiabatic gauge potentials

The central idea of counterdiabatic (CD) driving [10, 11] is to add an auxiliary Hamiltonian  $H_{cd}(t)$  to the original  $H_0(t)$  such that the total evolution under  $H(t) = H_0(t) + H_{cd}(t)$  exactly tracks the instantaneous eigenstate of  $H_0(t)$ , even when the adiabatic condition is violated. The exact CD term [11] requires full spectral knowledge of  $H_0(t)$  at all times and typically involves highly nonlocal interactions, making it impractical.

S. V. Romero *et al.* [13] employ the variational approach to approximate CD driving developed by D. Sels and A. Polkovnikov [21]. The approximate CD Hamiltonian at order  $l$  takes the form

$$H_{cd}^{(l)} = \dot{\lambda} \mathcal{A}_\lambda^{(l)}, \quad (24)$$

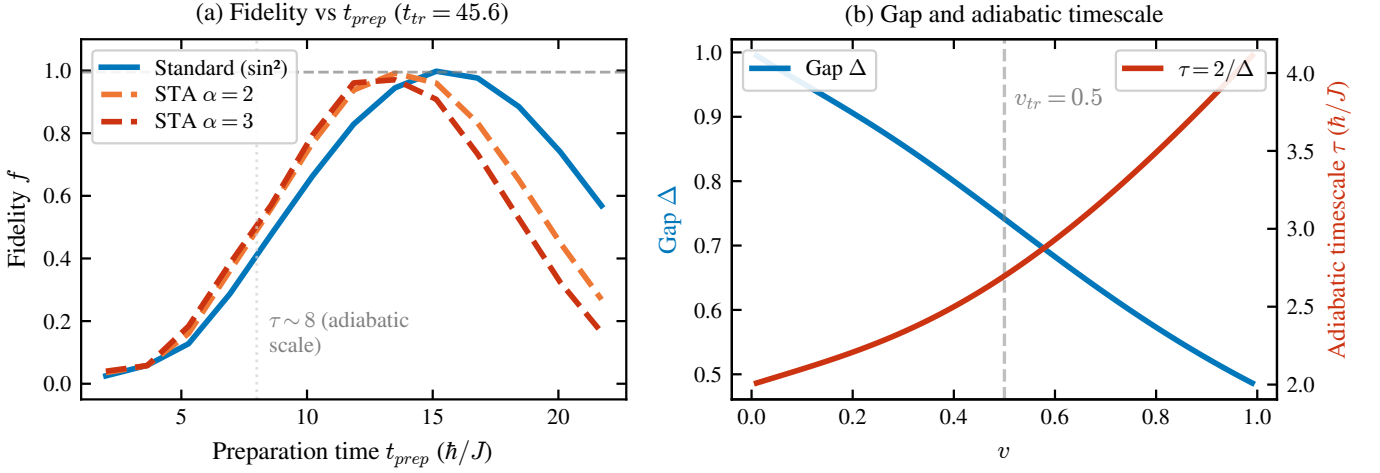


FIG. 11. Optimal transfer fidelity as a function of preparation time  $t_{\text{prep}}$  for different pulse profiles. The fidelity saturates once  $t_{\text{prep}}$  exceeds the adiabatic timescale  $\tau \sim 2/\Delta_{\text{min}}$ .

where  $\dot{\lambda} = \dot{t}_1(t)$  is the rate of change of the control parameter and  $\mathcal{A}_\lambda^{(l)}$  is the approximate adiabatic gauge potential, expanded in terms of nested commutators (NCs) of  $H_0$  and  $\partial_\lambda H_0$ :

$$\mathcal{A}_\lambda^{(l)} = i \sum_{k=1}^l \alpha_k^{(l)} \underbrace{[H_0, [H_0, \dots, [H_0, \partial_\lambda H_0]]]}_{2k-1}. \quad (25)$$

The time-dependent coefficients  $\alpha_k^{(l)}(t)$  are determined variationally by minimizing the action  $S_l = \text{tr}[G_l^2]$ , where  $G_l = \partial_\lambda H_0 - i[H_0, \mathcal{A}_\lambda^{(l)}]$ . In the limit  $l \rightarrow \infty$ , the expansion converges to the exact gauge potential. For finite  $l$ , one obtains an approximate CD driving whose accuracy improves with order.

### C. First-order nested commutator

The first-order ( $l = 1$ ) gauge potential involves only the single commutator  $[H_0, \partial_\lambda H_0]$ , yielding [13]

$$\mathcal{A}_\lambda^{(1)} = i\kappa^{(1)} \left[ \sum_{j=1}^{N-2} c_{2j+2}^\dagger c_{2j} - \sum_{j=1}^{N-1} c_{2j+1}^\dagger c_{2j-1} \right] + \text{H.c.}, \quad (26)$$

where  $\kappa^{(1)} = -\Lambda \dot{\lambda} \alpha_1^{(1)}$  with  $\Lambda = 2t_0$ . The variational parameter is obtained analytically [13, 21] as

$$\alpha_1^{(1)} = -\frac{\mathcal{C}(N)}{t_1^2 + t_2^2}, \quad \mathcal{C}(N) = \frac{2(N-2)+1}{8(N-2)+1}. \quad (27)$$

The physical interpretation of Eq. (26) is highly significant: *the first-order CD driving consists of next-to-nearest-neighbor (NNN) hopping terms between sites of the same sublattice—odd-odd (sublattice A) and even-even (sublattice B) with equal magnitude  $|\kappa^{(1)}|$ . Since the zero-energy eigenstate [Eq. (21)] has zero amplitude*

on all sublattice B sites, the NNN hoppings on sublattice B do not contribute to the transfer and can be neglected. This simplification reduces the CD driving to NNN hoppings exclusively between neighboring sublattice A sites (Fig. 13), which is far more experimentally tractable than the generic nonlocal interactions of the exact CD Hamiltonian (18).

### D. Higher-order nested commutators

The first-order NC provides excellent results for short chains (five sites) but becomes insufficient for longer chains, where higher-order corrections are needed. The second-order NC ( $l = 2$ ) introduces additional terms:

$$\mathcal{A}_\lambda^{(2)} = i(\alpha_1^{(2)}[H_0, \partial_\lambda H_0] + \alpha_2^{(2)}[H_0, [H_0, [H_0, \partial_\lambda H_0]]]), \quad (28)$$

which, after evaluation, produces NNN hoppings (as from the first order) plus fourth-nearest-neighbor hoppings within and between sublattices. Again, the sublattice B terms can be neglected for the zero-energy state transfer.

In general, the  $l$ th-order NC introduces interactions up to  $2l$ th-nearest neighbors [13]. For a chain with  $2N - 1$  sites, the minimum order required to account for all possible interactions along the same sublattice is  $l = N - 1$ . This rapid growth of computational complexity and long-range interactions motivates two key simplifications performed in Ref. [13]:

1. **Neglect sublattice B hoppings:** Since the transfer trajectory follows the zero-energy eigenstate, which has zero amplitude on sublattice B, the CD hoppings between even-even sites do not contribute.
2. **Retain only NNN hoppings on sublattice A:** The longer-range hoppings (fourth-nearest neigh-



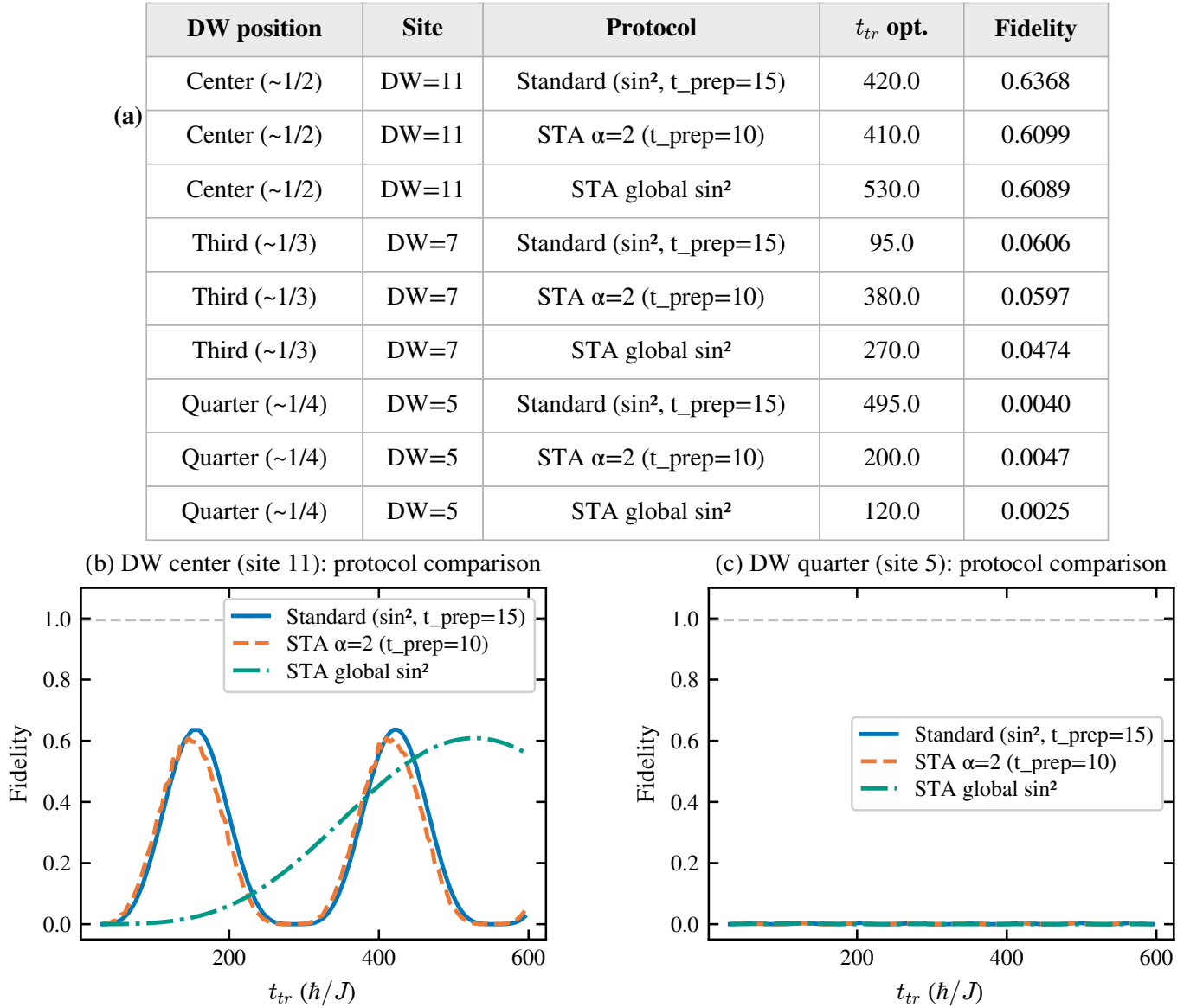


FIG. 12. Combined effect of asymmetry and STA pulses on the transfer fidelity ( $L = 21$ ). No pulse optimization can overcome the fundamental coupling asymmetry that limits the maximum achievable fidelity.

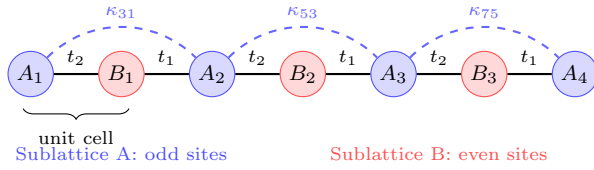


FIG. 13. Schematic of the hybrid analog-digital protocol for a chain of  $2N - 1 = 7$  sites. The original SSH hopping parameters  $t_1(t)$ ,  $t_2(t)$  (solid lines) are supplemented by counterdiabatic NNN hoppings  $\kappa_{2j+1,2j-1}$  (dashed arcs) connecting neighboring sublattice A sites. The NNN terms between sublattice B sites are negligible for the zero-energy eigenstate transfer.

bors and beyond) from higher-order NCs are discarded, keeping only the dominant NNN terms. The simplified CD driving takes the form

$$H_{cd} = -i \sum_{j=1}^{N-1} \kappa_{2j+1,2j-1} c_{2j+1}^\dagger c_{2j-1} + \text{H.c.}, \quad (29)$$

where  $\kappa_{2j+1,2j-1}$  are site-dependent coupling strengths to be optimized.

This simplified form retains the essential physics of the CD correction while requiring only experimentally accessible NNN hoppings. For a five-site chain, this simplified protocol achieves fidelities indistinguishable from the full second-order NC, confirming that the NNN sublattice A terms capture the dominant CD contribution.

### E. Analog counterdiabatic results

For a five-site chain ( $N = 3$ ) in the nonadiabatic regime ( $\Omega = t_0$ ,  $T = \pi$ ), the performance of the different CD approximations is [13]:

- **No CD** ( $H_0$  only): The fidelity drops significantly in the nonadiabatic regime, falling below 50% for  $\Omega \gtrsim 0.1 t_0$ .
- **First-order NC** ( $H_0 + H_{\text{cd}}^{(1)}$ ): The transfer improves substantially but does not reach high fidelity for  $\Omega = t_0$ .
- **Second-order NC** ( $H_0 + H_{\text{cd}}^{(2)}$ ): The protocol retains high fidelity  $F(T) > 99\%$  even at  $\Omega = t_0$  (transfer  $100\times$  faster than the adiabatic regime), with CD driving strengths of the same order as  $t_0$ .

These results demonstrate that a modest order of the NC expansion suffices for short chains. However, for chains longer than five sites, higher orders become necessary, and the analytical approach becomes computationally expensive. This motivates the variational digital optimization described next.

### F. Variational quantum circuit optimization

To circumvent the limitations of the analytical NC approach, S. V. Romero *et al.* [13] leverage digital quantum simulation (DQS) to find optimal CD driving parameters. The key idea is to treat the NNN hopping strengths  $\kappa_{2j+1,2j-1}$  as variational parameters and optimize them using classical optimizers in a hybrid quantum-classical loop [22].

#### 1. Encoding and Trotterization

The SSH chain of  $2N - 1$  sites is encoded in  $n = \lceil \log_2(2N - 1) \rceil$  qubits. Both  $H_0$  and  $H_{\text{cd}}$  are decomposed in the Pauli basis, and the time evolution is implemented via first-order Trotter-Suzuki decomposition [23]:

$$U(T, 0) \approx \prod_{k=1}^r e^{-iH_0^{(c)}(k\Delta t)\Delta t} e^{-iH_{\text{cd}}^{(c)}(k\Delta t)\Delta t}, \quad (30)$$

with  $r = T/\Delta t$  Trotter steps. The NNN coupling strengths are discretized as a vector of  $r + 1$  parameters per hopping location, with boundary conditions  $\kappa(0) = \kappa(T) = 0$ , leaving  $r - 1$  free parameters per location to optimize.

#### 2. Cost functions

Two cost functions are employed [13]:

1. **Transfer fidelity:** Directly maximizing  $C(\vec{\kappa}^{(c)}) = F(T)$  [Eq. (23)], which requires computing state overlaps and is well-suited for noiseless simulators.

2. **Hellinger distance:** A measurement-based cost function

$$H^2(P, Q) = 1 - \sum_{i=0}^{m-1} \sqrt{p_i q_i}, \quad (31)$$

where  $P$  is the measured probability distribution after  $N_{\text{shot}}$  measurements and  $Q$  is the target distribution (unit probability at the final site). Minimizing  $H^2$  is equivalent to maximizing the population at the target site and is practical for real quantum hardware [13].

The optimization uses the simultaneous perturbation stochastic approximation (SPSA) algorithm, initialized with values proportional to the analytical first-order NC result  $\kappa^{(1)}$  (“warm starting”) and with adaptive bounds to keep  $\|H_{\text{cd}}\| \sim \|H_0\|$  for experimental feasibility.

#### 3. Two simplification strategies

Two simplification ansatzes are tested for the NNN hoppings:

1. **Time-symmetric hoppings:** Exploiting the symmetry  $\kappa_{2i+1,2i-1}^{(c)}(t) = \kappa_{2(N-i)+1,2(N-i)-1}^{(c)}(T - t)$ , which halves the number of free parameters to  $\lfloor N/2 \rfloor (r - 1)$ .

2. **Equal NNN hoppings:** All NNN couplings are set equal,  $\kappa_{2j+1,2j-1}^{(c)}(t) \mapsto \kappa^{(c)}(t)$ , reducing the optimization to only  $r - 1$  parameters regardless of chain length. This is inspired by the site-independent structure of the first-order NC.

#### 4. Results across chain lengths

The variational circuit is implemented with  $r = 22$  Trotter steps, with 10 independent runs per configuration to assess stability. Table VI summarizes the optimal fidelities obtained across chain lengths from 5 to 15 sites [13].

For short chains (5–9 sites), the optimized CD driving achieves near-perfect fidelities with modest optimization effort. For longer chains (11–15 sites), the fidelity decreases to approximately 90% when the constraint  $\|H_{\text{cd}}\| \sim \|H_0\|$  is maintained. Relaxing this constraint (allowing stronger NNN hoppings) improves the fidelity at the expense of experimental feasibility.

### G. Transfer solely by counterdiabatic driving

A remarkable finding of Ref. [13] is that high-fidelity transfer can be achieved by setting  $H_0 = 0$  and driving the system entirely through the CD terms. In this regime, the circuit contains only the variational NNN hoppings on sublattice A, which simplifies the implementation in several ways:

- The circuit depth is significantly reduced (no  $H_0$  block),
- Convergence is faster, requiring only 1000 iterations for all chain lengths tested,
- The fidelities consistently exceed 99% for chains up to 15 sites, outperforming the  $H_0 + H_{cd}$  scheme.

This counterintuitive result can be understood as follows: when  $H_0 = 0$ , the population is entirely confined to sublattice A (no excitation of sublattice B sites), and the variational optimizer has full freedom to design the transfer pathway without competition from the original Hamiltonian's dynamics. The optimal CD driving in this case effectively encodes all the information about the SSH topology into a purely sublattice A hopping protocol.

### H. Robustness against disorder

The robustness of the CD protocol is assessed against three types of perturbations [13]:

1. **Rice-Mele staggered potential:** A diagonal term  $\sum_j (-1)^{j-1} \Delta(t) c_j^\dagger c_j$  that breaks chiral symmetry. The fidelity remains above 90% for  $\delta/t_0 \lesssim 0.1$ .
2. **Diagonal disorder:** Random on-site energies  $\delta_j \in [-\sigma_\delta, \sigma_\delta]$ . High fidelity is preserved for  $\sigma_\delta/t_0 \lesssim 0.1$ .
3. **Off-diagonal (hopping) disorder:** Random perturbations to the hopping amplitudes  $\tau_j \in [-\sigma_\tau, \sigma_\tau]$ . The protocol maintains high fidelity for  $\sigma_\tau/t_0 \lesssim 0.1$ .

TABLE VI. Transfer fidelities  $F(T)$  from the variational quantum circuit optimization [13] for different chain lengths in the nonadiabatic regime ( $\Omega = t_0$ ,  $T = \pi$ ), using equal NNN hoppings on sublattice A with the transfer fidelity as cost function.

| Sites | $N$ | Qubits | Iterations | $\bar{F}(T)$ |
|-------|-----|--------|------------|--------------|
| 5     | 3   | 3      | 1000       | $> 0.999$    |
| 7     | 4   | 3      | 2000       | $> 0.99$     |
| 9     | 5   | 4      | 3000       | $> 0.99$     |
| 11    | 6   | 4      | 5000       | $\sim 0.97$  |
| 13    | 7   | 4      | 6000       | $\sim 0.94$  |
| 15    | 8   | 4      | 10000      | $\sim 0.90$  |

The CD protocol with different (site-dependent) NNN hoppings is generally more robust than the equal-hopping version, as the site-dependent parameters can partially compensate local perturbations. The robustness threshold of  $\sim 10\%$  disorder strength relative to  $t_0$  is consistent with the energy scale hierarchy required for the CD correction to remain a perturbative supplement.

## VII. COMPARISON OF TRANSFER PARADIGMS

Having analyzed both the domain wall protocol (Secs. III–V) and the hybrid analog-digital CD protocol (Sec. VI), we now provide a systematic comparison of both approaches.

### A. Physical mechanism

The two protocols exploit fundamentally different physical mechanisms for edge-state transfer:

**Domain wall approach** (J. Zurita *et al.* [9]). The chain contains topological domain walls that create protected boundary states ( $|\mathcal{L}\rangle$ ,  $|\mathcal{S}\rangle$ ,  $|\mathcal{R}\rangle$ ) within the bulk gap. The system is prepared in one boundary state and the hopping amplitude  $v(t)$  is ramped adiabatically to populate the target state at the opposite edge. The transfer operates within a *low-energy subspace* spanned by the topologically protected zero-energy states, and the bulk gap provides protection against excitations. The domain wall itself is a static structural feature of the chain.

**CD approach** (S. V. Romero *et al.* [13]). The chain is a uniform SSH model (no domain wall). The hoppings  $t_1(t)$ ,  $t_2(t)$  are swept dynamically, and the transfer occurs by following the instantaneous zero-energy eigenstate of  $H_0(t)$  through the topological phase diagram. To accelerate this process without losing fidelity, auxiliary NNN hoppings are added as counterdiabatic corrections that suppress nonadiabatic transitions to other eigenstates. The transfer mechanism is a *dynamical shortcut to adiabaticity* rather than a topological subspace approach.

### B. Chain structure

A key practical difference concerns the chain topology:

- The **domain wall protocol** requires careful engineering of the chain structure, placing domain walls at specific positions. As demonstrated in Sec. IV, even small deviations from the ideal wall position cause exponential degradation of the transfer fidelity [Eq. (17)]. This imposes stringent fabrication requirements.
- The **CD protocol** operates on a uniform SSH chain with no structural defects. The complexity

is shifted from the chain architecture to the control fields: time-dependent NNN hoppings  $\kappa(t)$  must be engineered between sublattice A sites. These can be tuned continuously during the protocol.

### C. Scalability

Both approaches face scalability challenges, but of different nature:

**Domain wall protocol.** The transfer time scales exponentially with the domain length [Eq. (16)]:  $t_{\text{tr}} \sim (w/v)^{\ell/2}$ . For  $w = 2v$  and  $\ell = 10$ ,  $t_{\text{tr}} \sim 2^5 = 32$  (in units of  $1/v$ ). The multi-domain construction of Ref. [9] mitigates this by dividing the chain into  $N$  shorter domains, reducing the transfer time from  $(w/v)^{L/2}$  to  $N \times (w/v)^{L/(2N)}$ —an exponential improvement. The protocol has been demonstrated numerically for chains up to  $L \sim 100$  sites [9], though the number of domain walls must increase proportionally.

**CD protocol.** The operation time is fixed at  $T = \pi/\Omega$  (nonadiabatic regime), which does *not* scale with the chain length. However, the fidelity degrades for longer chains: the approximate CD correction (NNN hoppings only) becomes less accurate as the chain length increases, since higher-order NCs—which introduce increasingly long-range hoppings—are needed but are truncated. With the equal-hopping ansatz and the constraint  $\|H_{\text{cd}}\| \sim \|H_0\|$ , the fidelity drops to  $\sim 90\%$  for a 15-site chain [13]. The  $H_0 = 0$  variant maintains  $F > 99\%$  for 15 sites but requires engineering a purely CD-driven system without the reference Hamiltonian.

Table VII provides a quantitative comparison.

### D. Experimental requirements

The domain wall protocol requires only nearest-neighbor hopping control with a single time-dependent parameter  $v(t)$ . The domain wall structure is *static* and must be encoded at fabrication time. This makes it naturally suited to platforms with fixed lattice geometry, such as quantum dot arrays [24] or photonic lattices [25].

The CD protocol additionally requires time-dependent NNN hopping terms  $\kappa(t)$  between sublattice A sites. While more complex to control, this is feasible in several platforms: in superconducting qubit chains, NNN hoppings can be implemented via geometric phases or artificial gauge fields [13]; in cold atoms, angular momentum states [13] or synthetic dimensions provide tunable long-range couplings. The digital optimization via variational circuits provides discretized solutions that are naturally implementable in gate-based quantum processors.

### E. Robustness comparison

The two protocols exhibit different robustness characteristics:

- The domain wall protocol enjoys *topological protection*: the boundary states are robust against any perturbation that respects chiral symmetry. However, perturbations that break chiral symmetry (such as on-site disorder or Rice-Mele potentials) can mix the protected subspace with bulk states. Furthermore, as demonstrated in Sec. IV, the protocol is exquisitely sensitive to the *structural* asymmetry of the domain wall position—a form of “geometric disorder” that no pulse optimization can correct.
- The CD protocol is robust against moderate disorder ( $\sigma/t_0 \lesssim 0.1$ ) without relying on topological protection. The robustness arises from the variational optimization, which learns CD driving profiles that are effective even in the presence of perturbations. However, since the protocol deliberately breaks chiral symmetry by introducing NNN hoppings, it does not benefit from topological protection in the conventional sense.

### F. Complementarity of the two approaches

The two paradigms are complementary rather than competitors. The domain wall protocol excels in scenarios where:

- The chain structure can be fabricated with high precision,
- Only nearest-neighbor couplings are available,
- Chiral-symmetry-preserving conditions can be maintained,
- Long chains are needed (via multi-domain construction).

The CD protocol is preferable when:

- The chain structure cannot be precisely controlled,
- NNN couplings are experimentally accessible,
- Fast operation (nonadiabatic regime) is required to beat decoherence,
- A uniform chain is simpler to fabricate than one with domain walls.

A natural future direction is to combine both approaches: using domain wall-mediated transfer as the primary mechanism while adding CD corrections to suppress the residual nonadiabatic excitations during the preparation and readout ramps. This “best of both

TABLE VII. Comparison of the two transfer protocols. DW = domain wall, CD = counterdiabatic.

| Feature                     | DW protocol [9]                                   | CD protocol [13]                                               |
|-----------------------------|---------------------------------------------------|----------------------------------------------------------------|
| Chain structure             | Domain walls at specific sites                    | Uniform SSH chain                                              |
| Transfer mechanism          | Rabi oscillations in topological subspace         | Following instantaneous eigenstate via CD                      |
| Interactions                | Nearest-neighbor only                             | NN + NNN (sublattice A)                                        |
| Number of sites             | $L$ odd or even, $N \geq 2$ domains               | $2N - 1$ sites (odd)                                           |
| Transfer time               | $t_{\text{tr}} \sim (w/v)^{\ell/2}$ , exponential | $T = \pi/\Omega$ , fixed (nonadiabatic)                        |
| Scalability (time)          | Exponential; mitigated by multi-domain            | Constant time; fidelity decreases with $L$                     |
| Fidelity ( $L = 11$ , sym.) | $f = 0.999$                                       | $F > 0.97$ ( $H_0 + H_{\text{cd}}$ ); $F > 0.99$ ( $H_0 = 0$ ) |
| Sensitivity                 | Exponential to DW position                        | No structural sensitivity                                      |
| Disorder robustness         | Topological protection                            | Robust for $\sigma/t_0 \lesssim 0.1$                           |
| Control complexity          | Single ramp $v(t)$                                | Time-dependent NNN $\kappa(t)$ profiles                        |
| Optimization                | $t_{\text{prep}}, t_{\text{tr}}$                  | $r(N-1)$ or $r$ parameters                                     |
| Experimental platforms      | QD, SC qubits, cold atoms                         | Same; requires NNN coupling tunability                         |

worlds” strategy would leverage topological protection for the transfer phase while using CD driving for the time-critical ramp phases—precisely the regime where the STA protocols of Sec. V were shown to provide only marginal improvements.

### VIII. CTAP: DYNAMIC CONTROL BEYOND THE STATIC RABI REGIME

The results of Sec. V established that *no ramp-based pulse optimization* can overcome the fidelity degradation caused by coupling asymmetry in asymmetric domain wall configurations. However, this finding must be interpreted carefully. It does not imply that topological protection fails for asymmetric chains: the three edge states ( $|\mathcal{L}\rangle$ ,  $|\mathcal{S}\rangle$ ,  $|\mathcal{R}\rangle$ ) remain perfectly well-defined and well-separated from the bulk in all configurations studied. What fails is the specific *transfer protocol*—a static Rabi oscillation scheme that requires equal couplings  $J_{\mathcal{LS}} = J_{\mathcal{SR}}$ .

In this section we show, following the CTAP (Coherent Tunnelling Adiabatic Passage) framework of A. D. Greentree, J. H. Cole, A. R. Hamilton, and L. C. L. Hollenberg [14], that high-fidelity transfer in asymmetric SSH domain wall systems can be recovered by switching from the static Rabi protocol to a dynamic counter-intuitive pulse sequence. The key insight is that the effective three-level Hamiltonian *always* possesses a zero-energy dark state that mediates direct  $|\mathcal{L}\rangle \rightarrow |\mathcal{R}\rangle$  transfer without populating  $|\mathcal{S}\rangle$ , regardless of the magnitude of  $J_{\mathcal{LS}}$  and  $J_{\mathcal{SR}}$ .

#### A. Reframing the asymmetry constraint

Our results demonstrate that the fidelity collapse in asymmetric configurations is *not* a fundamental limitation of topological protection. It is an artifact of the static Rabi protocol: the  $\sin^2$  ramp fixes the hopping  $v(t)$  uniformly across the entire chain, so both  $J_{\mathcal{LS}}$  and  $J_{\mathcal{SR}}$

are determined by the same  $v_{\text{tr}}$  parameter. In asymmetric configurations ( $\ell_1 \neq \ell_2$ ), this leads to an exponentially large coupling ratio  $J_{\mathcal{LS}}/J_{\mathcal{SR}} \sim (w/v)^{(\ell_2 - \ell_1)/2}$  [Eq. (12)], for which the Rabi maximum fidelity is bounded by Eq. (17).

The topological protection is unaffected: the edge states are exponentially localized, the gap remains open, and the effective Hamiltonian still has the three-site tight-binding form (9). The asymmetry introduces no instability in the protected subspace—it merely makes the Rabi oscillation amplitude small. The natural solution is therefore to *decouple the control of the left and right domains*, enabling a dynamic protocol in which the coupling ratio  $J_{\mathcal{LS}}(t)/J_{\mathcal{SR}}(t)$  is varied in time to exploit the dark state of the effective Hamiltonian.

#### B. The effective three-level Hamiltonian and its dark state

We now allow the effective couplings to be time-dependent by introducing independent control parameters  $v_L(t)$  and  $v_R(t)$  for the intracell hopping in the left and right domains, respectively. The time-dependent effective Hamiltonian in the protected subspace reads

$$\mathcal{H}_{\text{eff}}(t) = \begin{pmatrix} 0 & J_{\mathcal{LS}}(t) & 0 \\ J_{\mathcal{LS}}(t) & 0 & J_{\mathcal{SR}}(t) \\ 0 & J_{\mathcal{SR}}(t) & 0 \end{pmatrix} \quad (32)$$

in the basis ( $|\mathcal{L}\rangle, |\mathcal{S}\rangle, |\mathcal{R}\rangle$ ), where both couplings now depend on time through the independent domain controls. This is directly analogous to the three-well Hamiltonian of Ref. [14] [their Eq. (1)], with our  $J_{\mathcal{LS}}(t)$  and  $J_{\mathcal{SR}}(t)$  playing the roles of the tunneling rates  $\Omega_{12}(t)$  and  $\Omega_{23}(t)$ , respectively.

The eigenvalues of Eq. (32) are

$$E_{\pm} = \pm \sqrt{J_{\mathcal{LS}}^2(t) + J_{\mathcal{SR}}^2(t)}, \quad E_0 = 0, \quad (33)$$

so the middle eigenvalue is *always zero*, independent of the magnitudes and ratio of the couplings. The corre-

sponding zero-energy eigenstate (the *dark state*) is [14]

$$|D_0(t)\rangle = \cos \Theta(t) |\mathcal{L}\rangle - \sin \Theta(t) |\mathcal{R}\rangle, \quad (34)$$

where the *mixing angle*  $\Theta(t)$  is defined by

$$\tan \Theta(t) = \frac{J_{\mathcal{L}\mathcal{S}}(t)}{J_{\mathcal{S}\mathcal{R}}(t)}. \quad (35)$$

The crucial property of  $|D_0(t)\rangle$  is that it has *zero component on  $|\mathcal{S}\rangle$* : the soliton state at the domain wall is completely absent from the dark state. This is a direct consequence of the chiral structure of the effective three-site chain and is preserved at all times regardless of the coupling ratio.

The dark state provides a natural transfer mechanism: if at  $t = 0$  we have  $J_{\mathcal{L}\mathcal{S}} = 0$  and  $J_{\mathcal{S}\mathcal{R}} \neq 0$ , then  $\Theta = 0$  and  $|D_0\rangle = |\mathcal{L}\rangle$  (the particle is on the left). If at  $t = T$  we have  $J_{\mathcal{L}\mathcal{S}} \neq 0$  and  $J_{\mathcal{S}\mathcal{R}} = 0$ , then  $\Theta = \pi/2$  and  $|D_0\rangle = -|\mathcal{R}\rangle$  (the particle is on the right). A slow, adiabatic change of  $\Theta(t)$  from 0 to  $\pi/2$  transfers the population from  $|\mathcal{L}\rangle$  to  $-\lvert\mathcal{R}\rangle$  with perfect fidelity, *without ever populating  $|\mathcal{S}\rangle$* .

### C. Counter-intuitive pulse sequence

The practical implementation of the dark-state transfer follows the counter-intuitive pulse ordering introduced by A. D. Greentree *et al.* [14] and originally developed in the context of optical STIRAP (Stimulated Raman Adiabatic Passage) [12]. The counter-intuitive aspect is that the *right-domain* coupling  $J_{\mathcal{S}\mathcal{R}}(t)$  must be switched on *before* the left-domain coupling  $J_{\mathcal{L}\mathcal{S}}(t)$ —the opposite of the intuitive ordering.

To see why this is necessary, note that for the system to start in  $|D_0\rangle = |\mathcal{L}\rangle$  at  $t = 0$ , we need  $\Theta(0) = 0$ , which requires  $J_{\mathcal{L}\mathcal{S}}(0) = 0$ . Similarly,  $|D_0(T)\rangle = -|\mathcal{R}\rangle$  requires  $\Theta(T) = \pi/2$ , meaning  $J_{\mathcal{S}\mathcal{R}}(T) = 0$ . So  $J_{\mathcal{S}\mathcal{R}}$  must be nonzero first (to define the dark state as  $|\mathcal{L}\rangle$ ) and must go to zero last (when the dark state becomes  $-\lvert\mathcal{R}\rangle$ ).

Concretely, we implement the two couplings using Gaussian pulse profiles analogous to Eq. (7) of Ref. [14]:

$$J_{\mathcal{S}\mathcal{R}}(t) = J_{\max}^R \exp\left[-\frac{(t - t_R)^2}{2\sigma^2}\right], \quad (36)$$

$$J_{\mathcal{L}\mathcal{S}}(t) = J_{\max}^L \exp\left[-\frac{(t - t_L)^2}{2\sigma^2}\right], \quad (37)$$

where  $t_R < t_L$  (the right pulse fires first) and  $t_L - t_R = \sigma$  (the pulse separation equals the standard deviation for optimal adiabaticity). In the SSH model,  $J_{\mathcal{S}\mathcal{R}}(t)$  is controlled by varying  $v_R(t)$  in the right domain, and  $J_{\mathcal{L}\mathcal{S}}(t)$  is controlled by  $v_L(t)$  in the left domain, both independently.

The critical point is that the peak amplitudes  $J_{\max}^R$  and  $J_{\max}^L$  *do not need to be equal*. For an asymmetric domain wall with  $\ell_1 \neq \ell_2$ , the coupling constants scale differently with  $v_L$  and  $v_R$  [Eq. (10)]. One can simply

tune the amplitude of each pulse independently to ensure the desired time-dependence of  $\Theta(t)$ . The adiabaticity condition  $|E_{\pm} - E_0| \gg |\langle \dot{D}_0 | D_{\pm} \rangle|$  reads [14]

$$\sqrt{J_{\mathcal{L}\mathcal{S}}^2 + J_{\mathcal{S}\mathcal{R}}^2} \gg \frac{|\dot{\Theta}|}{1} = \frac{|J_{\mathcal{S}\mathcal{R}}\dot{J}_{\mathcal{L}\mathcal{S}} - J_{\mathcal{L}\mathcal{S}}\dot{J}_{\mathcal{S}\mathcal{R}}|}{J_{\mathcal{L}\mathcal{S}}^2 + J_{\mathcal{S}\mathcal{R}}^2}, \quad (38)$$

which can always be satisfied by making the pulses sufficiently long, regardless of the coupling asymmetry.

### D. Counterdiabatic driving on the effective three-level system

The CTAP protocol is adiabatic and thus intrinsically slow: the pulse duration must satisfy the adiabaticity condition (38). To accelerate the protocol, one can apply the shortcut-to-adiabaticity strategy directly to the effective three-level Hamiltonian (32).

The exact counterdiabatic (CD) correction for a time-dependent two-level dark-state system can be derived following Berry's method [11, 12]. For the three-level system with a single mixing angle  $\Theta(t)$ , the result takes a particularly simple form [12]:

$$\mathcal{H}_{\text{CD}}^{\text{eff}}(t) = i\hbar \dot{\Theta}(t) (|\mathcal{L}\rangle\langle\mathcal{R}| - |\mathcal{R}\rangle\langle\mathcal{L}|), \quad (39)$$

where  $\dot{\Theta}(t) = (J_{\mathcal{S}\mathcal{R}}\dot{J}_{\mathcal{L}\mathcal{S}} - J_{\mathcal{L}\mathcal{S}}\dot{J}_{\mathcal{S}\mathcal{R}})/(J_{\mathcal{L}\mathcal{S}}^2 + J_{\mathcal{S}\mathcal{R}}^2)$  is the rate of change of the mixing angle. Adding  $\mathcal{H}_{\text{CD}}^{\text{eff}}$  to  $\mathcal{H}_{\text{eff}}(t)$  [Eq. (32)] suppresses all non-adiabatic transitions exactly, allowing perfect transfer in arbitrarily short times.

Physically,  $\mathcal{H}_{\text{CD}}^{\text{eff}}$  requires a *direct imaginary coupling* between  $|\mathcal{L}\rangle$  and  $|\mathcal{R}\rangle$ . In the microscopic SSH chain, this would correspond to engineering a long-range hopping between site 0 and site  $L - 1$  that is purely imaginary—a challenging but not impossible requirement in platforms with gauge-field control (e.g., via Floquet engineering or synthetic dimensions). Alternatively, the CD term can be approximated using local operators via the variational NC approach of Sec. VIB, applied now to the effective rather than the full Hamiltonian.

An important advantage of the CTAP+CD approach over the full-chain CD protocol (Sec. VI) is that here the CD term acts on the *three-dimensional* protected subspace rather than the  $(2N - 1)$ -dimensional full Hilbert space. This makes the optimization significantly more tractable and the required interactions more local (within the effective subspace).

### E. Advantages over the static Rabi protocol in asymmetric systems

The CTAP approach offers three decisive advantages over the static Rabi protocol in the presence of coupling asymmetry:

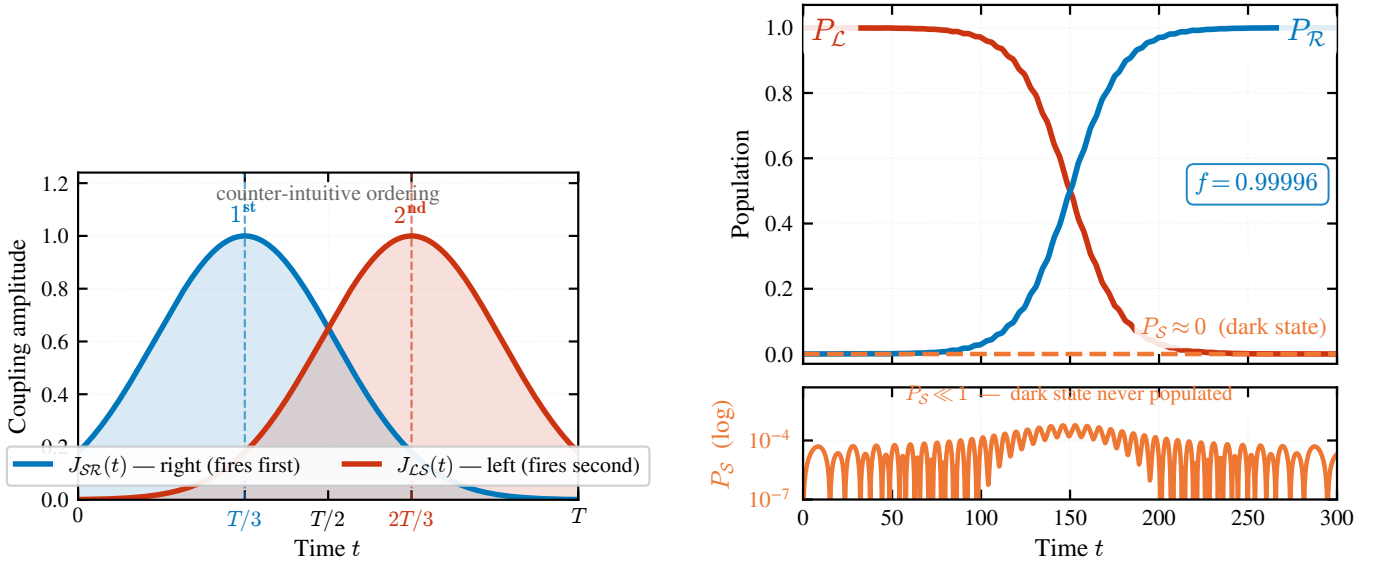


FIG. 14. **(a)** Counter-intuitive CTAP pulse sequence. The right-domain coupling  $J_{SR}(t)$  (blue) fires before the left-domain coupling  $J_{LS}(t)$  (red), adiabatically rotating the mixing angle  $\Theta(t)$  from  $0$  to  $\pi/2$  over the transfer time  $T$ . **(b)** Population dynamics of the effective three-level system under the CTAP protocol ( $T = 300$ ,  $\sigma = 0.18T$ ,  $J_{\max} = 1$ ). Starting from  $|\mathcal{L}\rangle$  (red), population is transferred to  $|\mathcal{R}\rangle$  (blue) with final fidelity  $f = 0.9999$ . The soliton state  $|\mathcal{S}\rangle$  (gold, dashed) remains at zero throughout (lower panel, log scale)—the hallmark of dark-state transfer via the topological zero mode.

- 1. Asymmetry invariance.** The fidelity of CTAP is independent of the ratio  $J_{LS}^{\max}/J_{SR}^{\max}$  as long as the adiabaticity condition is satisfied. Whether  $\ell_1 = \ell_2$  or  $\ell_1 \ll \ell_2$ , the protocol can be tuned to achieve  $f \rightarrow 1$  by appropriate choice of pulse duration. This is in stark contrast with the static Rabi protocol, for which  $f_{\max} \rightarrow 0$  exponentially as  $|\ell_1 - \ell_2| \rightarrow \infty$  [Eq. (17)].
- 2. No population of the soliton state.** Because the transfer occurs entirely through the dark state  $|D_0(t)\rangle$ , the soliton state  $|\mathcal{S}\rangle$  is never populated during the transfer. In the original Rabi protocol,  $|\mathcal{S}\rangle$  is transiently populated (up to  $1/2$  population at  $t = \pi/(2\sqrt{2}J)$ ), which can enhance decoherence in platforms where the domain wall site is more exposed to environmental noise.
- 3. Topological origin of the dark state.** The existence of the zero-energy dark state  $|D_0(t)\rangle$  in Eq. (34) is directly tied to the chiral symmetry of the effective Hamiltonian [Eq. (32)]. The latter is preserved as long as the SSH chain remains in the topological phase, independently of the domain wall position. CTAP thus combines the robustness of topological protection with the flexibility of dynamic control.

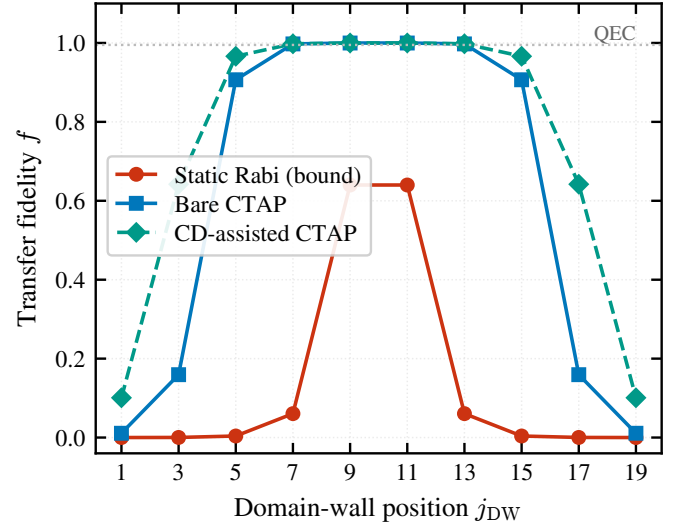


FIG. 15. Transfer fidelity versus domain-wall position  $j_{DW}$  for  $L = 21$  ( $v = 0.5$ ,  $w = 1.0$ ), obtained by numerical integration of the effective three-level Hamiltonian (32). The static Rabi protocol (red circles) peaks at  $f \approx 0.64$  near the quasi-centre and collapses to  $f \lesssim 0.004$  for  $j_{DW} \leq 5$ . Bare CTAP (blue squares), driven by the counter-intuitive Gaussian pulses [Eqs. (36)–(37)], sustains  $f > 0.99$  over a broad central region and degrades only for extreme asymmetry. CD-assisted CTAP (teal diamonds) widens the high-fidelity plateau by suppressing the residual nonadiabatic error. The dotted line marks the fault-tolerance threshold  $f_0 = 0.995$ .

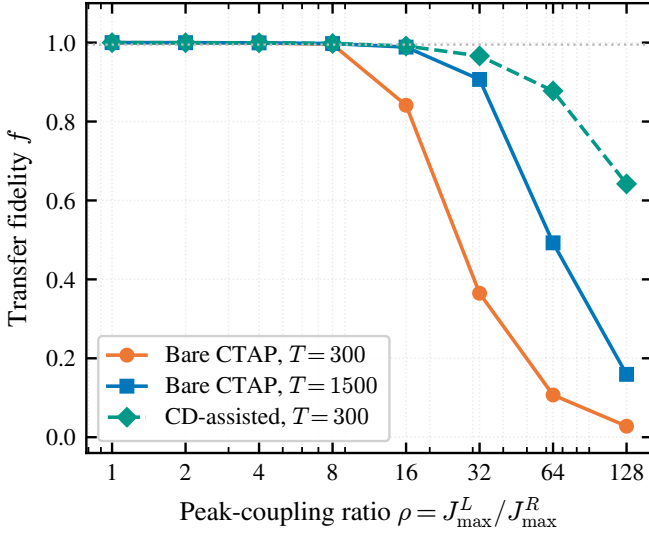


FIG. 16. Transfer fidelity as a function of the peak-coupling ratio  $\rho = J_{\max}^L / J_{\max}^R$  for the effective three-level system. Bare CTAP at  $T = 300$  (orange) and  $T = 1500$  (blue), and CD-assisted CTAP at  $T = 300$  (teal). High fidelity is maintained up to  $\rho \approx 16$ ; beyond  $\rho \sim 10^2$  the protected gap closes and the dark-state mechanism ceases to apply. Dotted line:  $f_0 = 0.995$ .

#### F. Numerical verification and the physical limit

To substantiate the asymmetry-resilience of CTAP we integrate the effective three-level Hamiltonian (32) with the counter-intuitive Gaussian pulses [Eqs. (36)–(37)] for every valid domain-wall position of the  $L = 21$  chain. The peak amplitude of the weaker pulse is set to  $1/\rho$ , where  $\rho = (w/v)^{|\ell_1 - \ell_2|/2}$  is the geometric coupling ratio that also bounds the Rabi protocol [Eq. (17)]; the longer domain physically caps its coupling because the intracell hopping must satisfy  $v < w$  to keep the bulk gap open.

Figure 15 reports the result. Bare CTAP recovers  $f > 0.99$  wherever  $\rho \lesssim 8$  ( $|\ell_1 - \ell_2| \lesssim 6$ ) and reaches  $f \approx 0.91$  at  $\rho = 32$  ( $j_{\text{DW}} = 5$ ) for sufficiently long pulses—more than two orders of magnitude above the Rabi bound  $f_{\max} = 0.004$ . Adding the counterdiabatic term (39) raises this to  $f \approx 0.97$ .

Figure 16 isolates the dependence on the peak ratio  $\rho$ . Both variants are essentially perfect for  $\rho \lesssim 8$ ; the bare protocol degrades earlier and is partially rescued by longer protocol times, while CD assistance sustains  $f > 0.99$  up to  $\rho \approx 16$ . For  $\rho \gtrsim 10^2$  all variants fall, because the weaker coupling  $J_{\min} \sim 1/\rho$  shrinks the instantaneous gap  $\sqrt{J_{\mathcal{LS}}^2 + J_{\mathcal{SR}}^2}$  until the three protected levels become quasi-degenerate and the effective three-level description—not the CTAP mechanism itself—breaks down. This is the genuine physical limit of dark-state transfer in a single-domain-wall geometry: CTAP removes the asymmetry bottleneck of the static Rabi protocol across the entire experimentally relevant range, and CD driving extends it further, but it cannot manufacture

coupling where the geometry forbids it.

#### G. Extension to two domain walls: even-state parity constraint

A natural extension of the domain wall transfer scheme is to consider chains with *two* domain walls, which support *four* topologically protected states:  $|\mathcal{L}\rangle$ ,  $|\mathcal{S}_1\rangle$ ,  $|\mathcal{S}_2\rangle$ , and  $|\mathcal{R}\rangle$ . The effective Hamiltonian in this subspace has the structure of a four-site tight-binding chain,

$$\mathcal{H}_{\text{eff}}^{(4)} = \begin{pmatrix} 0 & J_{12} & 0 & 0 \\ J_{12} & 0 & J_{23} & 0 \\ 0 & J_{23} & 0 & J_{34} \\ 0 & 0 & J_{34} & 0 \end{pmatrix}, \quad (40)$$

and one might hope that an analogous CTAP or straddling-CTAP (SCTAP) pulse sequence could transfer population from  $|\mathcal{L}\rangle$  to  $|\mathcal{R}\rangle$ .

However, A. D. Greentree *et al.* [14] identified a fundamental obstacle related to the *parity of the number of effective sites*: the CTAP scheme as described above is only effective for systems with an *odd* number of sites in the effective chain. For an even-site system (such as our four-state chain), there is no isolated zero-energy eigenstate; instead, there are two states whose energies approach zero but never vanish simultaneously (Fig. 6 of Ref. [14]).

Physically, the absence of an isolated dark state means that any adiabatic transfer trajectory must pass through a region of near-degeneracy between the two low-energy states. Satisfying the adiabaticity condition  $|E_0 - E_{\pm}| \gg |\langle \dot{D}_0 | D_{\pm} \rangle|$  in this region requires an *exponentially slow* protocol, negating the advantage of the CTAP approach.

To circumvent this parity constraint, A. D. Greentree *et al.* [14] introduced the *Straddling CTAP* (SCTAP) scheme, which employs an augmented Hamiltonian with three different pulse amplitudes:

$$\mathcal{H}_{\text{SCTAP}} = \begin{pmatrix} 0 & \Omega_1 & 0 & 0 \\ \Omega_1 & 0 & \Omega_S & 0 \\ 0 & \Omega_S & 0 & \Omega_2 \\ 0 & 0 & \Omega_2 & 0 \end{pmatrix}. \quad (41)$$

Here  $\Omega_1$  and  $\Omega_2$  are the counter-intuitive “outer” pulses (analogous to the CTAP  $J_{\mathcal{LS}}$  and  $J_{\mathcal{SR}}$ ), and  $\Omega_S$  is a persistent “straddling” pulse that remains on throughout the protocol. The zero-energy eigenstate of this Hamiltonian [Greentree Eq. (11)] has a modified form that still avoids populating the intermediate sites, but only in the limit  $X = \Omega_1 \Omega_2 / (\Omega_S \sqrt{\Omega_1^2 + \Omega_2^2}) \ll 1$ .

For a two-domain-wall SSH chain, the SCTAP approach would translate into:

1. A persistent nonzero hopping  $v_S$  coupling the two inner soliton states  $|\mathcal{S}_1\rangle$  and  $|\mathcal{S}_2\rangle$ , which can be implemented by leaving the central domain in a state of partial dimerization.



2. A counter-intuitive pair of outer pulses  $v_R(t)$  (right domain, fires first) and  $v_L(t)$  (left domain, fires second), analogous to the one-domain-wall CTAP.

A full analysis of this scheme, including the optimization of  $\Omega_S$  and verification of the  $X \ll 1$  condition for different chain geometries, is deferred to future work.

## IX. DISCUSSION AND CONCLUSIONS

We have presented a detailed study of quantum state transfer in SSH chains, comparing two distinct paradigms: domain wall-mediated transfer following J. Zurita *et al.* [9] and hybrid analog-digital counterdiabatic driving following S. V. Romero *et al.* [13], and proposing CTAP as the path beyond the limitations of the static Rabi regime.

Our main results can be summarized as follows:

1. **Faithful reproduction of the topological transfer protocol.** For the symmetric configuration ( $L = 11$ ,  $N = 2$ ,  $\ell = 4$ ), we reproduce the results of Ref. [9] with high precision: the optimal transfer time  $t_{\text{tr}} = 45$  agrees with the analytical prediction  $t_{\text{tr}} \approx 45.6$ , and the fidelity  $f = 0.999$  exceeds the fault-tolerance threshold.

2. **Exponential sensitivity to domain wall position.** Moving the domain wall away from the chain center degrades the fidelity according to the fundamental bound  $f_{\text{max}} = 4J_1^2 J_2^2 / (J_1^2 + J_2^2)^2$ . The coupling ratio  $J_1/J_2 \sim (w/v)^{|l_1 - l_2|/2}$  grows exponentially with the displacement, making the fidelity exquisitely sensitive to the wall position. For a single unit-cell displacement in  $L = 21$ , the fidelity drops from  $\sim 1$  to 0.637; for larger displacements, the transfer becomes negligible.

3. **Static Rabi protocol cannot overcome the coupling asymmetry.** Shortcut-to-adiabaticity pulse modifications provide marginal improvements in the symmetric case (the global  $\sin^2$  achieves  $f = 0.99995$ ) but are completely ineffective against the coupling asymmetry in the transfer phase. This establishes that the fidelity degradation in asymmetric configurations is *not a topological constraint* but an artifact of the static Rabi protocol. The topological edge states are fully intact; only the chosen transfer strategy is inadequate.

4. **Counterdiabatic driving as an alternative paradigm.** The hybrid analog-digital protocol of Ref. [13] circumvents the structural sensitivity of the domain wall approach by operating on a uniform SSH chain with time-dependent NNN hoppings as counterdiabatic corrections. In the nonadiabatic regime ( $T = \pi/\Omega$ , a  $100\times$  speed-up over the adiabatic protocol), the CD approach achieves fidelities above 99% for short chains (5–9 sites) and approximately 90% for 15 sites when CD

strengths are constrained to the same order as  $t_0$ .

5. **Complementarity of the two paradigms.** The domain wall and CD approaches are complementary: the former excels in platforms with precise structural control and nearest-neighbor-only couplings, while the latter is preferable for uniform chains with tunable NNN interactions and stringent time constraints.

6. **CTAP resolves the asymmetry problem via the topological dark state.** By introducing independent time-dependent controls for the left and right domains, the static Rabi limitation is overcome. The effective three-level Hamiltonian always possesses a zero-energy dark state  $|D_0(t)\rangle = \cos \Theta(t) |\mathcal{L}\rangle - \sin \Theta(t) |\mathcal{R}\rangle$ —a direct consequence of chiral symmetry—that mediates transfer from  $|\mathcal{L}\rangle$  to  $|\mathcal{R}\rangle$  without ever populating the soliton state  $|\mathcal{S}\rangle$ . The counter-intuitive CTAP pulse sequence ( $J_{SR}$  fires before  $J_{LS}$ ) adiabatically rotates  $\Theta(t)$  from 0 to  $\pi/2$ , achieving  $f > 0.99$  across the experimentally relevant range of coupling ratios (numerically verified up to  $\rho \approx 16$  with counterdiabatic assistance). The only residual limit is reached for extreme asymmetry ( $\rho \gtrsim 10^2$ ), where the weaker effective coupling shrinks the protected gap and the three-level description breaks down—a geometric, not a protocol, limitation.

7. **CD driving on the effective three-level system accelerates CTAP.** The counterdiabatic correction to the three-level effective Hamiltonian takes the form  $\mathcal{H}_{\text{CD}}^{\text{eff}} = i\hbar\dot{\Theta}(|\mathcal{L}\rangle\langle\mathcal{R}| - |\mathcal{R}\rangle\langle\mathcal{L}|)$ . Adding this term enables perfect population transfer in arbitrarily short times, combining the asymmetry resilience of CTAP with the speed of STA protocols. Since  $\mathcal{H}_{\text{CD}}^{\text{eff}}$  acts on a three-dimensional subspace, the variational optimization is far more tractable than for the full  $L$ -site chain.

8. **Even-state parity constraint limits two-domain-wall CTAP.** For chains with two domain walls, the four-dimensional protected subspace precludes a simple CTAP protocol due to the absence of an isolated zero-energy eigenstate for even numbers of effective sites. The Straddling CTAP (SCTAP) scheme of Ref. [14] provides a route forward, requiring a persistent coupling between the two inner soliton states and counter-intuitive outer pulses.

These findings provide a comprehensive picture of the state of the art in topological quantum state transfer via SSH chains. The exponential speed-up predicted for multi-domain ( $N > 2$ ) chains [9], the variational optimization strategies for CD driving [13, 26], the CTAP framework for asymmetry-resilient transfer [14], and the possibility of hybrid domain-wall/CD protocols offer multiple avenues for future investigation and experimental implementation across superconducting qubit, cold-atom, and quantum dot platforms.

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[1] S. Bose, Quantum communication through an unmodulated spin chain, *Physical Review Letters* **91**, 207901

(2003).

- [2] M. Christandl, N. Datta, A. Ekert, and A. J. Landahl, Perfect state transfer in quantum spin networks, *Physical Review Letters* **92**, 187902 (2004).
- [3] G. M. Nikolopoulos and I. Jex, *Quantum State Transfer and Network Engineering* (Springer, 2014).
- [4] M. Z. Hasan and C. L. Kane, Colloquium: Topological insulators, *Reviews of Modern Physics* **82**, 3045 (2010).
- [5] J. K. Asbóth, L. Oroszlány, and A. Pályi, *A Short Course on Topological Insulators: Band Structure and Edge States in One and Two Dimensions*, Lecture Notes in Physics, Vol. 919 (Springer, 2016).
- [6] W. P. Su, J. R. Schrieffer, and A. J. Heeger, Solitons in polyacetylene, *Physical Review Letters* **42**, 1698 (1979).
- [7] W. P. Su, J. R. Schrieffer, and A. J. Heeger, Soliton excitations in polyacetylene, *Physical Review B* **22**, 2099 (1980).
- [8] S. Ryu, A. P. Schnyder, A. Furusaki, and A. W. W. Ludwig, Topological insulators and superconductors: tenfold way and dimensional hierarchy, *New Journal of Physics* **12**, 065010 (2010).
- [9] J. Zurita, C. E. Creffield, and G. Platero, Fast quantum transfer mediated by topological domain walls, *Quantum* **7**, 1043 (2023).
- [10] M. Demirplak and S. A. Rice, Adiabatic population transfer with control fields, *The Journal of Physical Chemistry A* **107**, 9937 (2003).
- [11] M. V. Berry, Transitionless quantum driving, *Journal of Physics A: Mathematical and Theoretical* **42**, 365303 (2009).
- [12] D. Guéry-Odelin, A. Ruschhaupt, A. Kiely, E. Torrontegui, S. Martínez-Garaot, and J. G. Muga, Shortcuts to adiabaticity: Concepts, methods, and applications, *Reviews of Modern Physics* **91**, 045001 (2019).
- [13] S. V. Romero, X. Chen, G. Platero, and Y. Ban, Optimizing edge-state transfer in a Su-Schrieffer-Heeger chain via hybrid analog-digital strategies, *Physical Review Applied* **21**, 034033 (2024).
- [14] A. D. Greentree, J. H. Cole, A. R. Hamilton, and L. C. L. Hollenberg, Coherent electronic transfer in quantum dot systems using adiabatic passage, *Physical Review B* **70**, 235317 (2004), arXiv:cond-mat/0407008.
- [15] J. Zak, Berry's phase for energy bands in solids, *Physical Review Letters* **62**, 2747 (1989).
- [16] E. Prodan and H. Schulz-Baldes, *Bulk and Boundary Invariants for Complex Topological Insulators*, Mathematical Physics Studies (Springer, 2016).
- [17] E. Knill, Quantum computing with realistically noisy devices, *Nature* **434**, 39 (2005).
- [18] X. Chen, A. Ruschhaupt, S. Schmidt, A. del Campo, D. Guéry-Odelin, and J. G. Muga, Fast optimal frictionless atom cooling in harmonic traps: Shortcut to adiabaticity, *Physical Review Letters* **104**, 063002 (2010).
- [19] D. J. Thouless, Quantization of particle transport, *Physical Review B* **27**, 6083 (1983).
- [20] S. Longhi, Topological pumping of edge states via adiabatic passage, *Physical Review B* **99**, 155150 (2019).
- [21] D. Sels and A. Polkovnikov, Minimizing irreversible losses in quantum systems by local counterdiabatic driving, *Proceedings of the National Academy of Sciences* **114**, E3909 (2017).
- [22] M. Cerezo, A. Arrasmith, R. Babbush, S. C. Benjamin, S. Endo, K. Fujii, J. R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P. J. Coles, Variational quantum algorithms, *Nature Reviews Physics* **3**, 625 (2021).
- [23] H. F. Trotter, On the product of semi-groups of operators, *Proceedings of the American Mathematical Society* **10**, 545 (1959).
- [24] F. Mei, G. Chen, L. Tian, S.-L. Zhu, and S. Jia, Robust quantum state transfer via topological edge states in superconducting qubit chains, *Physical Review A* **98**, 012331 (2018).
- [25] E. J. Meier, F. A. An, and B. Gadway, Observation of the topological soliton state in the Su-Schrieffer-Heeger model, *Nature Communications* **7**, 13986 (2016).
- [26] N. N. Hegade, K. Paul, Y. Ding, M. Sanz, F. Albarrán-Arriagada, E. Solano, and X. Chen, Shortcuts to adiabaticity in digitized adiabatic quantum computing, *Physical Review Applied* **15**, 024038 (2021).